

Military Technical Academy (MTA)



**PhD Proposal: Deep Learning-Based CT
Reconstruction Methods**

Information technology

Code number:

PhD Student: Duong Thi Thuy

Supervisor: Assoc. Prof. Tran Cao Truong

Hanoi, 10/2025

Contents

Abstract	3
1 Introduction	3
1.1 Problem Statement	3
1.2 Motivation	6
1.3 Research Objectives	8
1.4 Research Methodology	11
2 Literature Review	15
2.1 Background	15
2.1.1 Deep learning	15
2.1.2 Low-Dose CT Imaging and Noise Characteristics	19
2.1.3 Overview of Low-Dose CT Reconstruction	22
2.1.4 Conventional Model-Based Reconstruction Techniques	26
2.1.5 Image Quality Metrics for LDCT Reconstructionn . . .	29
2.2 Related Work on Deep Learning for LDCT Reconstruction . .	33
2.2.1 Deep Learning-Based Reconstruction Methods	33
2.2.2 CNN-Based LDCT Reconstruction Approaches	36
2.2.3 Dual-Domain and Physics-Informed Learning Frame- works	40
2.2.4 Self - Supervised and Transfer Learning Strategies . . .	41
2.2.5 Strategies Applications of LDCT Reconstruction in Lung Cancer Screening	42
3 Proposed Contributions and Project Plan	47
3.1 Proposed Contributions	47
3.2 Thesis Outline	49
3.3 Project Timeline	51
References	53

1 Introduction

1.1 Problem Statement

Computed Tomography (CT) has become one of the most important imaging tools in modern clinical practice, providing cross-sectional visualization of internal organs with high spatial and contrast resolution^{1,2}. Despite its diagnostic advantages, conventional CT scanning exposes patients to relatively high doses of ionizing radiation, raising safety concerns, particularly for those who require repeated imaging or for large-scale population screening^{3,4}. The ongoing global increase in CT utilization has intensified concerns regarding radiation-induced malignancy and long-term stochastic effects, emphasizing the urgent need for effective dose-reduction strategies without compromising diagnostic accuracy³.

In recent years, Low-Dose CT (LDCT) techniques have emerged as a clinically attractive alternative because they can substantially reduce radiation exposure—typically by 60–80 % compared with normal-dose CT (NDCT) while maintaining similar coverage and spatial resolution^{1,2,5}. However, this reduction in radiation dose inevitably decreases the number of detected X-ray photons, which worsens the signal-to-noise ratio and introduces quantum noise and streak artifacts into the reconstructed images. The resulting LDCT scans often appear grainy, with blurred edges and non-uniform Hounsfield Unit (HU) values, leading to degraded image quality and reduced diagnostic confidence^{2,3,5}.

Among the various applications of CT, LDCT has demonstrated particular value in the early detection of lung cancer. Numerous clinical trials, most notably the National Lung Screening Trial (NLST) and the NELSON study, have shown that regular LDCT screening significantly reduces lung cancer mortality by identifying tumors at earlier, more treatable stages⁶. Nevertheless, the diagnostic success of LDCT screening depends critically on the quality and reliability of the reconstructed images. Excessive noise, loss of subtle textural patterns, or misrepresentation of HU values can obscure small pulmonary nodules, ground-glass opacities, or part-solid lesions that are indicative of early malignancy. Consequently, achieving NDCT-like image quality is not merely a matter of improving visual appearance—it is essential for ensuring diagnostic accuracy, reproducibility of radiomic features, and the long-term consistency required for follow-up and longitudinal analysis. For lung cancer screening to be safe and effective, LDCT must strike a delicate balance between minimizing radiation dose and maintaining image quality adequate for both radiologists and AI-assisted diagnosis^{1,5}.

To improve the quality of LDCT images, two major methodological families have been developed. The first involves iterative reconstruction (IR) algorithms, which explicitly model the physics and noise statistics of CT acquisition to suppress noise while preserving structural information⁶. The second, more recent and rapidly advancing approach leverages deep learning (DL) models such as convolutional neural networks (CNNs), generative adversarial networks (GANs), transformers, and diffusion-based architectures⁷. These models learn to transform noisy LDCT images into NDCT-quality images by training on large paired datasets, where each LDCT slice is aligned with a corresponding NDCT ground truth. This paired supervision enables networks to optimize pixel-wise and perceptual loss functions, resulting in reconstructed images that are both visually appealing and quantitatively accurate. State-of-the-art DL models trained on benchmark datasets, including the Mayo Clinic AAPM Challenge and the LIDC-IDRI database, have achieved remarkable improvements in image quality, with reported PSNR and SSIM values approaching those of real NDCT scans⁸.

Despite these promising advances, directly transferring such pretrained models to Vietnam is not straightforward. Most existing LDCT reconstruction models have been developed using highly standardized datasets acquired from Western populations, where scanner types, reconstruction kernels, and radiation protocols are uniform^{5,9}. The learned parameters of these models inherently encode the imaging physics, detector response, and anatomical characteristics of the training population. When these models are applied to a different imaging domain—such as Vietnamese CT data—the mismatch in acquisition protocols, scanner technology, and patient anatomy results in degraded performance. Vietnamese hospitals use multi-vendor CT systems, including Siemens, Canon, GE, and Philips, each employing different reconstruction filters, dose modulation strategies, and calibration standards. Furthermore, patient body composition, tissue attenuation, and respiratory patterns in the Vietnamese population differ markedly from those of Western subjects, altering the underlying image statistics and noise distributions. As a result, foreign-trained models may produce visually “clean” reconstructions that are, in fact, quantitatively unreliable—oversmoothing anatomical details, hallucinating artificial textures, or distorting HU values. These discrepancies highlight the urgent need for reconstruction models that are trained or adapted using Vietnamese imaging data and that reflect the specific hardware and population characteristics of the local healthcare system.

The challenge of implementing foreign models in Vietnam is compounded by three structural limitations. First, there is a significant data limitation: most Vietnamese hospitals currently archive only NDCT scans

for diagnostic purposes. Real LDCT datasets paired with NDCT counterparts do not exist, and access to raw projection data (sinograms) is restricted by equipment vendors. This absence of paired data renders conventional supervised learning approaches for LDCT reconstruction impractical^{10,11}. Second, there exists considerable scanner and protocol heterogeneity across the country. The variation in image reconstruction kernels, detector geometry, and radiation parameters leads to inconsistent noise power spectra, HU calibration, and spatial resolution, forming a substantial domain gap between Vietnamese and foreign data^{5,12}. Third, ensuring clinical reliability remains a critical concern^{4,5,13}. Even when a deep learning model effectively removes noise, it may inadvertently smooth or alter small lesions, particularly faint ground-glass opacities, thereby compromising diagnostic accuracy. A reconstructed image must therefore not only appear visually clear but also preserve diagnostically relevant details and maintain consistent HU and radiomic measurements that radiologists rely upon for clinical decisions.

These challenges have important implications for Vietnam’s efforts to develop a safe and effective LDCT-based lung cancer screening program. For radiologists, reconstructed images must be diagnostically reliable—comparable to NDCT scans in terms of noise texture, edge sharpness, and HU accuracy—to ensure consistent and defensible interpretations across institutions. For computational applications, downstream AI models used for nodule detection and malignancy prediction require inputs with stable distributions across scanners and dose levels; otherwise, performance will deteriorate due to calibration drift and data inconsistency. Therefore, LDCT reconstruction must go beyond perceptual enhancement and focus on preserving clinical and quantitative fidelity¹.

From a methodological perspective, these limitations redefine the core research question. Without access to paired LDCT–NDCT data or sinograms, traditional supervised reconstruction and physics-based unrolled optimization methods are infeasible. At the same time, purely data-driven denoising risks overfitting to foreign priors and may fail to preserve fine pathological structures. Consequently, there is a pressing need for a new class of deep learning models that can be trained exclusively on NDCT data from Vietnamese hospitals, learn to simulate scanner-specific LDCT noise patterns in a physics-guided or self-supervised manner¹⁴, reconstruct NDCT-like images that preserve HU accuracy and essential lung structures, and ultimately support AI-based lung cancer screening that is robust across different scanners and clinical environments⁵.

Equally important is how reconstruction success is evaluated. Traditional image quality metrics such as PSNR and SSIM, while useful, are

insufficient for clinical validation^{3,13}. Reconstruction performance must also be assessed through HU bias and variance analysis, noise power spectrum (NPS) similarity, edge-preservation indices, and the stability of radiomic features across reconstructed and reference NDCT images³. Furthermore, diagnostic integrity should be verified through reader studies with Vietnamese radiologists to ensure that reconstructed LDCT scans neither obscure nor create lesions and that they support consistent Lung-RADS assessments^{1,5}. Only by combining quantitative evaluation with clinical validation can an LDCT reconstruction model be deemed suitable for integration into real-world screening workflows.

In summary, the core problem addressed in this research is the following: Given that only NDCT data from Vietnamese hospitals are currently available for training, how can we develop deep learning models that[?] reconstruct future LDCT scans into diagnostically reliable NDCT-like images for radiologist interpretation, and enable accurate, domain-robust lung cancer detection and risk prediction suitable for national deployment? Addressing this challenge requires bridging both the data availability gap—the lack of paired LDCT–NDCT or projection data—and the domain generalization gap, caused by multi-vendor scanner variation and local anatomical differences, while maintaining strict clinical fidelity in HU accuracy, radiomic stability, and lesion preservation. The outcome of this work aims to establish the technical foundation for safe, low-dose CT imaging and AI-assisted lung cancer screening tailored to Vietnam’s healthcare context^{5,9}.

1.2 Motivation

The motivation behind this research stems from the urgent need to balance two conflicting demands in medical imaging: reducing radiation exposure for patients while maintaining the diagnostic image quality required for accurate disease detection. In the context of lung cancer screening, this balance is particularly critical. Lung cancer remains the leading cause of cancer-related deaths in Vietnam, and its high mortality rate is largely due to late diagnosis[?]. The adoption of low-dose CT (LDCT) screening programs offers a promising strategy to identify cancers at an early, more treatable stage^{1,4}. However, for such programs to be clinically viable, the LDCT images used for screening must be of sufficient quality for radiologists and AI systems to make reliable diagnostic judgments. Achieving this level of quality requires reconstruction models capable of producing NDCT-like images that preserve the visibility of small pulmonary nodules and subtle parenchymal patterns even at substantially reduced radiation doses⁹.

Despite significant global progress in LDCT reconstruction using deep learning, most of these models are trained on foreign datasets and assume access to paired LDCT–NDCT images⁹. Such conditions are unrealistic in Vietnam, where hospitals currently store only NDCT scans and cannot export raw projection data due to vendor and privacy restrictions¹⁵. Moreover, the diversity of CT scanners across Vietnamese institutions and the anatomical characteristics of the Vietnamese population create a unique imaging environment that differs markedly from that of Western datasets. Models developed abroad often fail to generalize under these conditions, producing reconstructions that look clean but lack quantitative reliability for clinical interpretation. Therefore, there is a clear scientific and practical motivation to develop LDCT reconstruction methods that can be trained exclusively on NDCT data from Vietnamese hospitals, while remaining robust and clinically trustworthy when applied to future LDCT scans acquired locally.

To meet this challenge, the research is motivated by three complementary directions that together form a coherent technical framework. The first motivation is to enable self-sufficient model training under limited data conditions. Since no paired LDCT–NDCT data exist in Vietnam, the proposed Physics-Guided Simulation and Reconstruction Network (PGSR-Net) will introduce a novel self-supervised approach that simulates LDCT noise from NDCT data based on scanner-specific parameters and then learns to reconstruct NDCT-like images from these simulations. By embedding physics-based constraints into a deep learning architecture, PGSR-Net aims to reproduce realistic LDCT conditions and achieve high-fidelity reconstruction without relying on external datasets¹⁶. This approach not only addresses the data availability problem but also ensures that the reconstructed images align with the physical and radiological properties of Vietnamese CT scanners¹⁷.

The second motivation is to ensure that the reconstructed images are optimized for diagnostic relevance, not merely for numerical image quality. Traditional reconstruction networks tend to maximize metrics such as PSNR or SSIM, which may not correlate with clinical interpretability¹⁸. To bridge this gap, the proposed Lung-Region Reconstruction Transformer (LRTR-Net) focuses on preserving the structures that are most important for lung cancer detection, such as small nodules, vessels, and alveolar patterns¹⁶. By incorporating attention mechanisms guided by lung masks and multi-scale semantic features, LRTR-Net will prioritize reconstruction in clinically critical regions while maintaining accurate HU and texture relationships. This diagnosis-driven strategy ensures that the reconstructed NDCT-like images are not only visually clear but also preserve the subtle pathological information necessary for early cancer detection.

The third motivation arises from the need to integrate reconstruction with robust and generalizable lung cancer prediction. Even with high-quality NDCT-like images, AI-based diagnostic models must be able to operate reliably across different scanners, doses, and patient populations. To achieve this, the proposed Contrastive Cancer Prediction Network (CCP-Net) leverages contrastive learning to align feature representations between NDCT-like and NDCT domains¹⁷. This alignment encourages the model to learn domain-invariant features that remain consistent regardless of imaging noise or scanner variations. As a result, CCP-Net is expected to provide accurate predictions of lung cancer risk from reconstructed LDCT images, ensuring that AI-assisted screening remains stable and trustworthy in real clinical environments.

Beyond technical innovation, the broader motivation of this research lies in its potential clinical and societal impact. By enabling high-quality LDCT reconstruction and reliable AI-based diagnosis using only NDCT data, the proposed framework can significantly accelerate the deployment of national lung cancer screening programs in Vietnam. It will reduce patient radiation exposure¹, improve diagnostic confidence, and create a foundation for large-scale AI integration into radiology workflows¹⁹. Moreover, the methods developed in this work can be extended to other imaging tasks and diseases, contributing to a broader shift toward safe, data-efficient, and population-specific AI in medical imaging²⁰.

In summary, this research is driven by the need to overcome the data scarcity and domain mismatch challenges unique to Vietnam, while maintaining high standards of diagnostic accuracy and clinical reliability. By combining physics-informed modeling (PGSR-Net), diagnosis-driven reconstruction (LRTR-Net), and domain-robust prediction (CCP-Net), the study seeks to build a unified framework for LDCT reconstruction and lung cancer screening that is scientifically novel, technically sound, and practically transformative for the Vietnamese healthcare system^{1,5,20}.

1.3 Research Objectives

The overall objective of this research is to develop a comprehensive deep learning framework for low-dose CT (LDCT) reconstruction and lung cancer prediction, optimized for the imaging conditions, scanner diversity, and population characteristics of Vietnam. At present, hospitals in Vietnam mainly possess normal-dose CT (NDCT) datasets, with no access to paired LDCT–NDCT data or raw projection (sinogram) data. This situation presents a major obstacle for applying supervised learning methods

traditionally used in LDCT image restoration¹⁷. Therefore, the central goal of this research is to design new deep learning models that can be trained solely on existing NDCT data yet remain effective when reconstructing future LDCT scans into diagnostically reliable NDCT-like images and predicting lung cancer risk with high accuracy and domain robustness. The proposed framework aims to integrate three complementary research directions: (1) physics-guided LDCT simulation and reconstruction, (2) diagnosis-driven lung-region reconstruction, and (3) contrastive learning for domain-robust lung cancer prediction. Together, these directions will address the key challenges of data scarcity, scanner variability, and clinical reliability, laying the foundation for a safe and effective AI-assisted LDCT screening program in Vietnam.

Research Objective 1 (RO1): Physics-Guided Simulation and Reconstruction Network (PGSR-Net).

The first objective focuses on developing a physics-guided deep learning model capable of simulating LDCT noise from existing NDCT images and reconstructing NDCT-like images without relying on paired data¹⁶. This objective directly addresses the most fundamental limitation of Vietnamese medical imaging—the absence of LDCT datasets and projection-domain information for supervised model training. The proposed Physics-Guided Simulation and Reconstruction Network (PGSR-Net) will combine the advantages of physics-based noise modeling with the flexibility of data-driven learning. It will be designed to learn the statistical characteristics of scanner-dependent noise and dose-level variations by using metadata extracted from DICOM headers, such as tube voltage (kVp), current (mAs), and CT dose index (CTDIvol)¹⁶. Based on these parameters, the model will simulate LDCT-like inputs by injecting synthetic noise that follows Poisson-Gaussian statistics to approximate real photon degradation.

Once the LDCT simulation module is trained, a second sub-network will be used for reconstruction, mapping simulated LDCT images back to NDCT-like quality. The reconstruction process will be guided by a combination of physics-consistent constraints (e.g., Radon-domain consistency), HU calibration losses to maintain accurate tissue density, and edge-preservation terms to retain fine anatomical details. The two modules will be jointly optimized in an end-to-end fashion, enabling the model to generate realistic LDCT images and perform denoising simultaneously. In essence, PGSR-Net will learn to reproduce the physical process of low-dose imaging and to reverse its degradation effects without ever requiring paired LDCT–NDCT data. Through this mechanism, the model is expected to generate NDCT-like images that closely approximate clinical-quality scans, providing radiologists

with diagnostically reliable results and creating high-quality input data for the subsequent research objectives.

Research Objective 2 (RO2): Lung-Region Reconstruction Transformer (LRTR-Net)

The second objective builds upon the outcomes of RO1 and aims to refine LDCT reconstruction through a diagnosis-driven, lung-region-aware deep learning model. While PGSR-Net emphasizes accurate physical reconstruction and global noise suppression, it may still fail to prioritize the restoration of clinically significant structures within the lungs—particularly small pulmonary nodules and fine vascular patterns that are crucial for early cancer detection. To overcome this limitation, the proposed Lung-Region Reconstruction Transformer (LRTR-Net) will introduce a reconstruction strategy that explicitly focuses on the anatomical regions most relevant for diagnosis. The motivation is to align reconstruction objectives not merely with pixel-level image quality (such as PSNR or SSIM) but with diagnostic relevance, ensuring that reconstructed images faithfully preserve subtle morphological and textural cues associated with malignancy.

The LRTR-Net will employ a hybrid architecture that integrates convolutional layers for low-level feature extraction with transformer-based attention mechanisms for capturing global and contextual dependencies. To focus the model on the lung region, segmentation masks generated from the NDCT or PGSR-Net outputs will be used as region priors. Within these masks, a dual attention mechanism will operate: spatial attention to enhance local detail around nodules and small vessels, and semantic attention to maintain HU contrast consistency across lung parenchyma. The model will be trained under a multi-task learning paradigm, combining reconstruction losses with feature-consistency losses derived from pretrained nodule detection networks. This design ensures that the model learns to optimize image quality in a way that directly benefits clinical tasks. The reconstructed NDCT-like outputs produced by LRTR-Net are expected to exhibit superior edge definition, realistic HU distribution, and improved detectability of early-stage lung lesions, thus serving as a reliable foundation for automated or human-assisted cancer diagnosis.

Research Objective 3 (RO3): Contrastive Cancer Prediction Network (CCP-Net)

The third objective extends the work from reconstruction to disease prediction, addressing the problem of domain robustness in AI-based lung cancer detection¹⁸. Even if reconstruction quality is high, deep learning classifiers trained on NDCT or NDCT-like data can still fail to generalize across different scanners or dose conditions, due to discrepancies in feature

distributions. The aim of Contrastive Cancer Prediction Network (CCP-Net) is to learn domain-invariant feature representations that remain stable between NDCT and LDCT-like domains, enabling consistent lung cancer risk prediction even when applied to real LDCT data from Vietnamese hospitals.

To achieve this, CCP-Net will use a dual-encoder contrastive learning framework, inspired by recent advances in self-supervised and domain-adaptive learning. One encoder will process NDCT data, while the other will process NDCT-like images reconstructed by LRTR-Net. The two encoders will be trained jointly using a contrastive objective function that maximizes similarity between corresponding NDCT and NDCT-like features while minimizing similarity between non-corresponding pairs. This training strategy will force the model to learn robust latent representations that capture intrinsic pathology-related information rather than scanner-specific noise or texture patterns. Once the shared feature space is established, it will be fine-tuned for nodule-level and patient-level lung cancer classification tasks using available NDCT labels. The model will also integrate handcrafted radiomic features with deep embeddings to enhance interpretability and ensure clinical alignment. Ultimately, CCP-Net aims to deliver a prediction model that can operate accurately on reconstructed LDCT data, maintaining consistent diagnostic performance across different hospitals, scanners, and acquisition settings.

These three objectives are designed to form an integrated research framework, where each component builds upon the results of the previous one. RO1 establishes the data foundation by generating realistic LDCT simulations and NDCT-like reconstructions from NDCT-only datasets. RO2 refines these reconstructions to ensure that diagnostic information—especially within the lung region—is preserved with high fidelity. RO3 utilizes the reconstructed and diagnosis-driven NDCT-like images to train a robust AI system capable of accurate lung cancer prediction. Together, these components form a complete end-to-end pipeline: from LDCT simulation and reconstruction to clinically meaningful diagnosis.

1.4 Research Methodology

The overall methodology of this research is designed to progressively address the challenges identified in the problem statement and to achieve the three research objectives proposed earlier. The study will follow a multi-stage experimental and development framework that combines data collection, model exploration, algorithm design, and validation¹⁷. Each stage will build upon the outcomes of the previous one to ensure both technical feasibil-

ity and clinical relevance. The research process consists of four key phases: (1) dataset acquisition and characterization, (2) benchmarking of existing LDCT reconstruction and diagnosis models, (3) development of the three proposed models (PGSR-Net, LRTR-Net, CCP-Net), and (4) comprehensive evaluation and validation using both public and Vietnamese data.

Phase 1: Dataset Collection and Preparation

The first phase focuses on collecting and preparing the data necessary to support both model development and evaluation. Since no public paired LDCT–NDCT datasets are currently available in Vietnam, the research will begin by consolidating existing international open-access datasets that have been widely used in LDCT reconstruction research. These include the AAPM Mayo Clinic Low-Dose CT Grand Challenge dataset, which provides matched LDCT–NDCT image pairs, and the LIDC-IDRI database, which contains lung nodule annotations and diagnostic labels. These datasets will serve two purposes: to benchmark existing reconstruction models and to pretrain baseline networks under controlled conditions.

In parallel, efforts will be made to collect and curate Vietnamese CT data from collaborating hospitals. This dataset will initially consist of anonymized NDCT chest scans acquired under various scanning conditions and from different vendors (e.g., Siemens, GE, Canon, Philips). Metadata such as tube voltage (kVp), tube current (mAs), slice thickness, and reconstruction kernel will be extracted from DICOM headers for use in scanner-specific modeling. The data will be preprocessed to standardize voxel size, intensity normalization, and lung field segmentation. This Vietnamese NDCT collection will later serve as the core dataset for developing self-supervised and physics-guided learning strategies. Data curation will strictly follow ethical and privacy regulations, ensuring patient anonymity and institutional compliance. The combination of international and local data will thus provide a foundation for both cross-domain benchmarking and domain-specific model adaptation.

Phase 2: Benchmarking of Existing Methods

Before developing new architectures, it is essential to establish a baseline performance reference by implementing and evaluating several state-of-the-art LDCT reconstruction and denoising algorithms. Representative deep learning models, such as RED-CNN, WGAN-VGG, Du-GAN, SwinIR, and Diffusion-LDCT, will be reimplemented and trained on the public datasets to reproduce published results. These models will then be tested on Vietnamese NDCT data (with synthetic LDCT noise) to analyze how well they generalize across domains. Performance will be measured using conventional quantitative metrics—PSNR, SSIM, RMSE—as well as radiomic stability and HU

accuracy¹⁷.

This benchmarking phase serves two purposes. First, it provides empirical evidence of the domain gap between international and Vietnamese CT data, quantifying how much reconstruction performance deteriorates when foreign-trained models are applied to local scans. Second, it helps identify the strengths and weaknesses of existing model architectures, which will inform the design of the proposed networks in later stages. Alongside reconstruction algorithms, available lung cancer detection models (e.g., DeepLung, 3D Faster R-CNN, TransBTS) will also be tested on both NDCT and simulated LDCT data to assess the impact of image degradation on diagnostic performance. These benchmarking experiments will provide a comprehensive understanding of current limitations and establish a baseline for improvement.

Phase 3: Development of Proposed Models

The core of this research lies in developing and integrating three novel deep learning models—PGSR-Net, LRTR-Net, and CCP-Net—each addressing a distinct but complementary aspect of the LDCT reconstruction and lung cancer prediction pipeline.

The first model, PGSR-Net (Physics-Guided Simulation and Reconstruction Network), will be designed to generate realistic LDCT simulations from NDCT scans and reconstruct NDCT-like images using a self-supervised physics-informed framework. The development process will begin by modeling scanner-dependent noise characteristics based on metadata from Vietnamese CT machines. The noise simulation module will employ Poisson-Gaussian distributions and dose-specific variance profiles to mimic photon starvation effects. The reconstruction network will then be trained end-to-end to recover NDCT-like images, guided by HU consistency, edge preservation, and physical constraints embedded in Radon-space consistency loss. The outputs from PGSR-Net will not only provide NDCT-like reconstructions for clinical visualization but also serve as training inputs for the subsequent LRTR-Net.

Building on PGSR-Net, LRTR-Net (Lung-Region Reconstruction Transformer) will focus on enhancing diagnostic precision in the lung region. The model will combine convolutional and transformer components to capture both local texture and global structural dependencies. Lung masks will be automatically extracted to direct attention mechanisms toward diagnostically significant regions. The model will be trained under a multi-task paradigm that optimizes for both reconstruction quality and diagnostic feature preservation, incorporating feature losses derived from pretrained nodule detection networks. Through this approach, LRTR-Net will generate NDCT-like im-

ages optimized for early lesion visibility and radiomic feature stability—an essential step for accurate AI-assisted diagnosis.

The third model, CCP-Net (Contrastive Cancer Prediction Network), will complete the pipeline by learning domain-invariant representations for lung cancer prediction. The model will use a dual-encoder contrastive framework, with one encoder trained on NDCT data and the other on NDCT-like images produced by LRTR-Net. Using contrastive learning objectives, CCP-Net will align latent features across domains, ensuring that diagnostic representations are insensitive to dose variation, scanner differences, and noise artifacts[?]. The model will then be fine-tuned for malignancy classification using annotated nodules from the LIDC-IDRI and Vietnamese datasets. The integration of radiomic and deep features will further enhance model interpretability and clinical robustness. Together, the three models will form a closed-loop system that spans from low-dose reconstruction to reliable disease prediction.

Phase 4: Evaluation and Validation

The final phase of the research involves rigorous quantitative and clinical evaluation of the proposed framework. Quantitative assessment will be performed using a variety of image quality metrics, including PSNR, SSIM, RMSE, HU bias, and noise power spectrum similarity. Radiomic reproducibility will be measured using intra-class correlation coefficients (ICC) between NDCT and reconstructed NDCT-like images¹⁷. The effectiveness of each model in preserving clinically relevant structures will be validated through region-specific metrics such as edge-preservation index and structural fidelity score. For diagnostic performance, CCP-Net will be evaluated on both public and Vietnamese datasets using receiver operating characteristic (ROC) analysis, area under the curve (AUC), precision–recall metrics, and calibration curves.

In addition to quantitative metrics, reader studies will be conducted in collaboration with Vietnamese radiologists to assess the perceptual and diagnostic quality of reconstructed images. Radiologists will evaluate the clarity of lesions, noise perception, and confidence in Lung-RADS classification when using NDCT-like images generated by the proposed models. The goal is to verify that the reconstructed LDCT images are not only visually improved but also clinically acceptable and trustworthy. Cross-site validation across different Vietnamese hospitals will further ensure that the models are robust to scanner and protocol variability[?].

2 Literature Review

2.1 Background

2.1.1 Deep learning

Computed Tomography (CT) is one of the most transformative innovations in diagnostic radiology, providing high-resolution cross-sectional and volumetric visualization of internal anatomy¹. Its fundamental principle is based on acquiring a series of X-ray projections from multiple angular positions around the patient and reconstructing them into cross-sectional images using mathematical algorithms such as filtered back projection (FBP) or iterative reconstruction (IR)²¹. Since its introduction by Hounsfield and Cormack in the 1970s, CT has evolved from early single-slice scanners to modern multi-detector and cone-beam systems^{1,2}, achieving isotropic spatial resolution (≤ 0.5 mm) and acquisition speeds below one second per rotation¹.

This technology revolutionized clinical diagnosis across oncology, cardiology, neurology, and trauma imaging by enabling accurate localization and quantification of pathological lesions. However, the principal limitation of CT is its reliance on ionizing radiation, which introduces potential biological risks such as radiation-induced carcinogenesis³. As the global use of CT continues to increase, dose management has become a critical issue in both clinical practice and research.

Radiation Exposure and Biological Risk

The radiation dose delivered during CT examinations is typically expressed as the effective dose (E, in millisieverts, mSv), which represents the weighted sum of organ-specific doses accounting for biological sensitivity. Typical diagnostic CT doses range between 2–20 mSv, approximately 100–1,000 chest radiographs². A standard chest CT delivers around 7 mSv, whereas an abdominal CT may reach 10–15 mSv.

Principles and Strategies for Dose Optimization

Modern CT systems employ a multi-pronged approach to balance diagnostic image quality with radiation safety. These strategies can be broadly classified into: (1) Acquisition parameter optimization, (2) Hardware and detector innovation, and (3) Algorithmic and reconstruction advances.

(a) Acquisition Parameter Optimization

Automatic Tube Current Modulation (ATCM) and Adaptive Voltage Control (AVC) dynamically adjust the tube current (mA) and tube voltage (kVp) in real time, based on patient size, anatomy, and attenuation properties. This approach reduces dose by 20–40% while preserving diag-

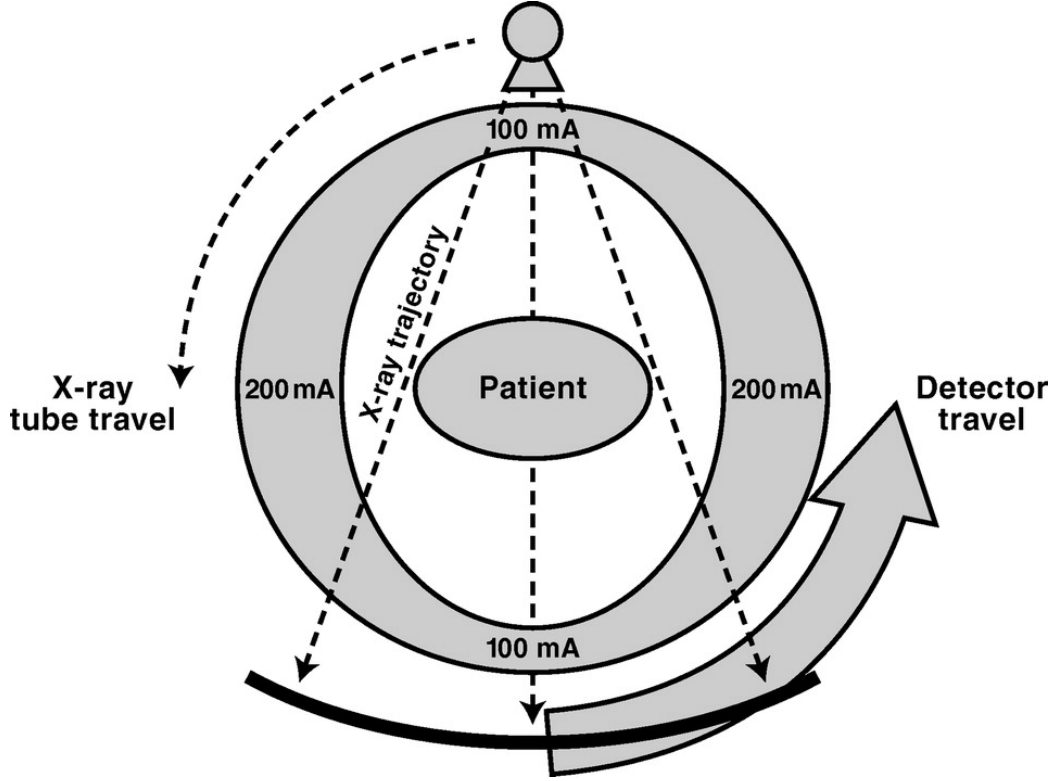


Figure 1. Basic schematic of CT imaging showing the rotating X-ray source, detectors, and reconstruction process.

nostic quality². Low-kVp scanning protocols (e.g., 80–100 kVp) are particularly beneficial in soft-tissue imaging and angiography, where they enhance contrast-to-noise ratio (CNR) and allow concurrent reductions in both radiation and contrast agent dose.

(b) Hardware and Detector Innovation

Photon-counting detector CT (PCD-CT) is a major hardware advancement that directly converts individual X-ray photons into electrical signals and classifies them by energy level. Compared with conventional energy-integrating detectors, PCDs reduce electronic noise and improve dose efficiency, achieving up to 40% dose reduction while maintaining image quality²². Dual-energy and spectral CT technologies further enable virtual non-contrast imaging and tissue decomposition, minimizing multiphase acquisitions²³

(c) Algorithmic and Reconstruction Advances

Reconstruction algorithms are pivotal to dose reduction. The traditional FBP method, though computationally efficient, amplifies noise under low-photon conditions²⁴. Iterative Reconstruction (IR) techniques, such as

Adaptive Statistical Iterative Reconstruction (ASIR-V) and Model-Based Iterative Reconstruction (MBIR), incorporate statistical modeling of photon noise and system physics, allowing 50–70% dose reduction while maintaining diagnostic confidence^{6,21?}.

Recent advances in Deep Learning Reconstruction (DLR) have further transformed dose optimization. Deep neural networks learn nonlinear mappings between low-dose and standard-dose images, suppressing noise and artifacts while preserving structural details. Clinical studies demonstrate that DLR achieves equivalent or superior image quality at up to 75% dose reduction compared with IR methods²⁵. Moreover, AI-based control systems are being developed to adaptively tune scanning parameters in real time, enabling patient-specific dose regulation.

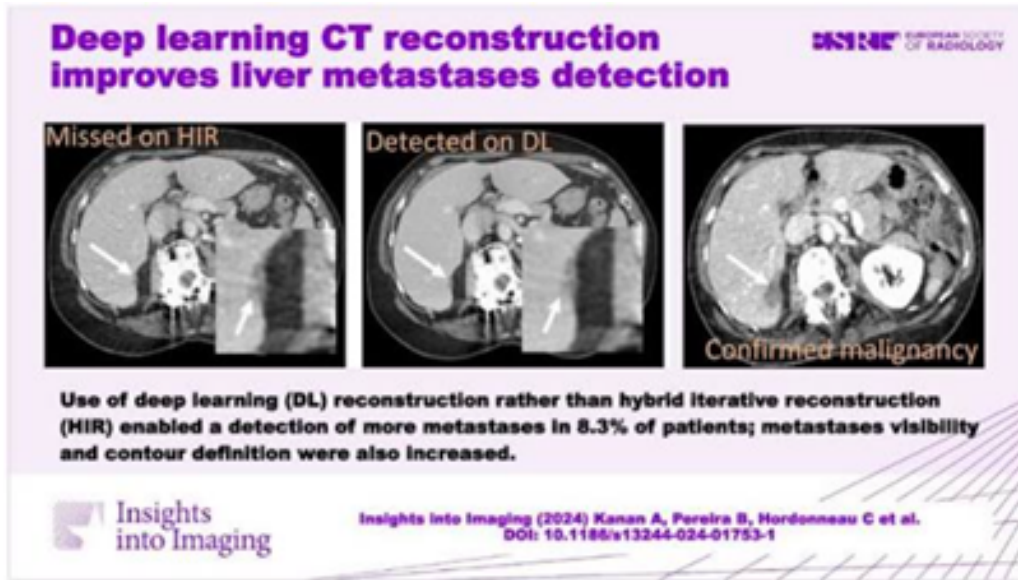


Figure 2. Comparison of image quality among FBP, IR, and DLR in abdominal low-dose CT. Deep learning reconstruction suppresses noise and preserves fine structures.

Clinical Implementation and Quantitative Evidence

Recent studies confirm the practical value of deep learning-based reconstruction in clinical settings:

- Jo et al. (2023) demonstrated approximately 75 % dose reduction (median 0.16 mSv vs 0.65 mSv) in ultra-low-dose chest CT using deep learning reconstruction, maintaining pulmonary nodule detection accuracy²⁵
- Tachibana et al. (2024) showed that combining 80 kVp acquisition with DLR reduced both radiation and iodine contrast dose by more than 50 % in abdominal CT while preserving diagnostic performance²⁶.

- Pozzessere et al. (2023) analyzed radiation exposure in low-dose CT lung cancer screening, reporting a 45–55 % reduction in effective dose and proportional decrease in estimated lifetime attributable cancer risk²⁷.

Challenges and Future Perspectives

Despite remarkable progress, several challenges persist:

- Model generalizability across scanner types and patient anatomies remains limited.
- Ultra-low-dose acquisitions (≤ 0.2 mSv) still suffer from photon starvation, degrading SNR and low-contrast detectability.
- Lack of standardized evaluation metrics for perceptual image quality hinders consistent validation.

Emerging physics-informed deep learning frameworks integrate system models and data priors, enhancing robustness under sparse or noisy acquisition conditions. These hybrid approaches are expected to enable sub-millisievert CT imaging with clinically reliable quality, advancing the next generation of AI-driven dose optimization.

Technique	Dose Reduction	Strengths	Limitations	Representative Studies
FBP	Baseline	Fast, simple	High noise, streak artifacts	[17], [18]
IR (ASIR-V, MBIR)	50–70%	Improved SNR, artifact reduction	Slow computation	[23], [24]
DLR	70–75%	Excellent noise suppression, edge preservation	Limited generalizability	[25]–[27]

Figure 3. *Comparison of CT Reconstruction Techniques for Dose Reduction*

Computed Tomography remains an indispensable diagnostic tool, yet radiation exposure management is an ongoing challenge. The evolution from hardware-based optimization to advanced iterative and deep learning-based reconstruction marks a paradigm shift toward intelligent, patient-specific dose control. Integration of physics-informed AI algorithms promises a new era of safe, ultra-efficient CT imaging that maximizes diagnostic confidence while adhering to the ALARA principle.

2.1.2 Low-Dose CT Imaging and Noise Characteristics

Low-Dose Computed Tomography (LDCT) has emerged as a pivotal innovation in diagnostic imaging, primarily driven by the need to minimize ionizing radiation exposure without compromising diagnostic image quality. Conventional CT scans typically involve radiation doses ranging from 5 to 15 mSv, depending on the anatomical region and imaging protocol²⁸. In contrast, LDCT seeks to reduce this exposure to approximately 1–2 mSv, or even below 1 mSv in specific screening applications such as lung cancer detection²⁹. The motivation behind this reduction is the well-established correlation between cumulative radiation dose and the stochastic risk of radiation-induced malignancies, as summarized in the linear no-threshold (LNT) model proposed by the International Commission on Radiological Protection (ICRP)³⁰.

However, dose reduction in CT imaging is intrinsically associated with the degradation of image quality, primarily due to the increased dominance of quantum noise. The number of detected photons decreases approximately linearly with the radiation dose, resulting in a higher relative variance of the detected signal. This phenomenon, commonly referred to as quantum mottle, manifests as grainy or speckled patterns in reconstructed CT images³¹. Figure 2.2 illustrates the visual difference between standard-dose and low-dose CT images, highlighting the increased noise level and the associated loss of structural detail.

Noise in LDCT images arises mainly from three sources: (i) statistical noise in photon detection (Poisson-distributed), (ii) electronic noise from the detector circuitry, and (iii) model-based errors in reconstruction algorithms. Among these, photon noise is the dominant component, especially at reduced tube currents (measured in mA). The overall noise level, typically quantified by the standard deviation of Hounsfield Units (HU) within homogeneous regions, increases approximately proportionally to the inverse square root of the dose, i.e., $\sigma \propto \frac{1}{\sqrt{D}}$ ³². Consequently, a 75% reduction in dose can lead to nearly a twofold increase in noise variance.

In the frequency domain, LDCT noise is characterized by a higher proportion of high-frequency components, contributing to the granular texture that impedes the visibility of subtle anatomical structures and lesions³³. The Signal-to-Noise Ratio (SNR) and Contrast-to-Noise Ratio (CNR) are two primary quantitative metrics used to evaluate image quality in LDCT reconstruction. SNR, defined as the mean pixel intensity divided by the standard deviation of background noise, typically decreases by 40–60% in LDCT compared to standard-dose CT³⁴. Similarly, CNR—measuring the contrast between two regions of interest relative to the background noise—declines significantly, particularly in soft tissue imaging, where the intrinsic contrast

is low.

To counteract these challenges, various acquisition techniques and reconstruction algorithms have been introduced to optimize the trade-off between radiation dose and image quality. These include tube current modulation, adaptive statistical iterative reconstruction (ASIR), and model-based iterative reconstruction (MBIR). Despite their effectiveness in suppressing noise, traditional methods are computationally intensive and may introduce oversmoothing, resulting in the loss of fine details essential for accurate diagnosis.

Recent research trends emphasize data-driven and deep learning-based denoising strategies, which learn the statistical characteristics of noise directly from large datasets. These methods demonstrate superior performance in restoring textural details and preserving diagnostic information, achieving improvements in Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) by 3–5 dB and 0.05–0.1, respectively, compared to conventional iterative techniques. Nevertheless, ensuring the generalizability of these methods across diverse scanners, protocols, and patient populations remains a key research challenge.

In addition to perceptual noise, LDCT images often exhibit non-stationary noise characteristics, meaning that the noise variance is spatially dependent. Regions of high attenuation (e.g., bone or contrast-filled vessels) tend to have higher photon absorption and thus lower SNR, whereas low-attenuation regions (e.g., lungs) exhibit relatively lower noise levels³⁵. This spatial heterogeneity complicates the task of uniform noise suppression and necessitates adaptive filtering or spatially variant regularization techniques in image reconstruction.

Quantitative assessment of LDCT image quality often employs phantom studies, where uniform objects are scanned at varying doses to determine noise magnitude, SNR, and modulation transfer function (MTF). Empirical studies have shown that reducing the tube current–time product (mAs) by 50% results in a mean SNR drop from approximately 25 to 12, accompanied by a PSNR decrease from 40 dB to around 33 dB³⁴. Such findings underscore the importance of sophisticated reconstruction algorithms that can compensate for the noise amplification inherent in low-dose imaging.

Ultimately, the pursuit of low-dose CT imaging represents a critical balance between radiation safety and diagnostic fidelity. Advances in hardware (such as photon-counting detectors) and software (particularly deep learning reconstruction frameworks) continue to reshape this balance, promising a future where sub-millisievert CT scans may become the norm for screening and follow-up examinations³⁶. Figure 2.3 schematically depicts the relationship

between dose reduction and image quality metrics, illustrating the trade-off and potential compensation achievable through advanced reconstruction techniques.

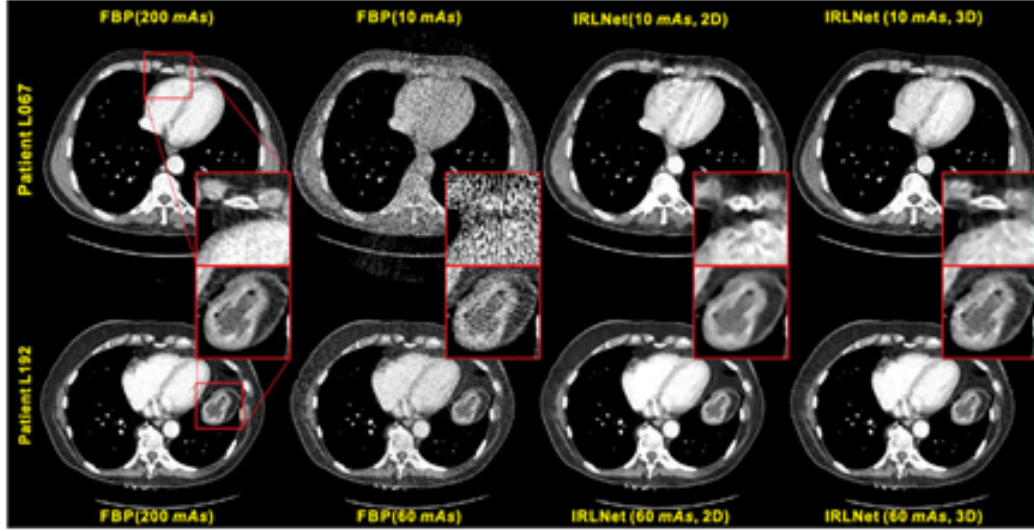


Figure 4. Comparison between standard-dose and low-dose CT images, demonstrating increased noise and structural blurring at reduced exposure levels.

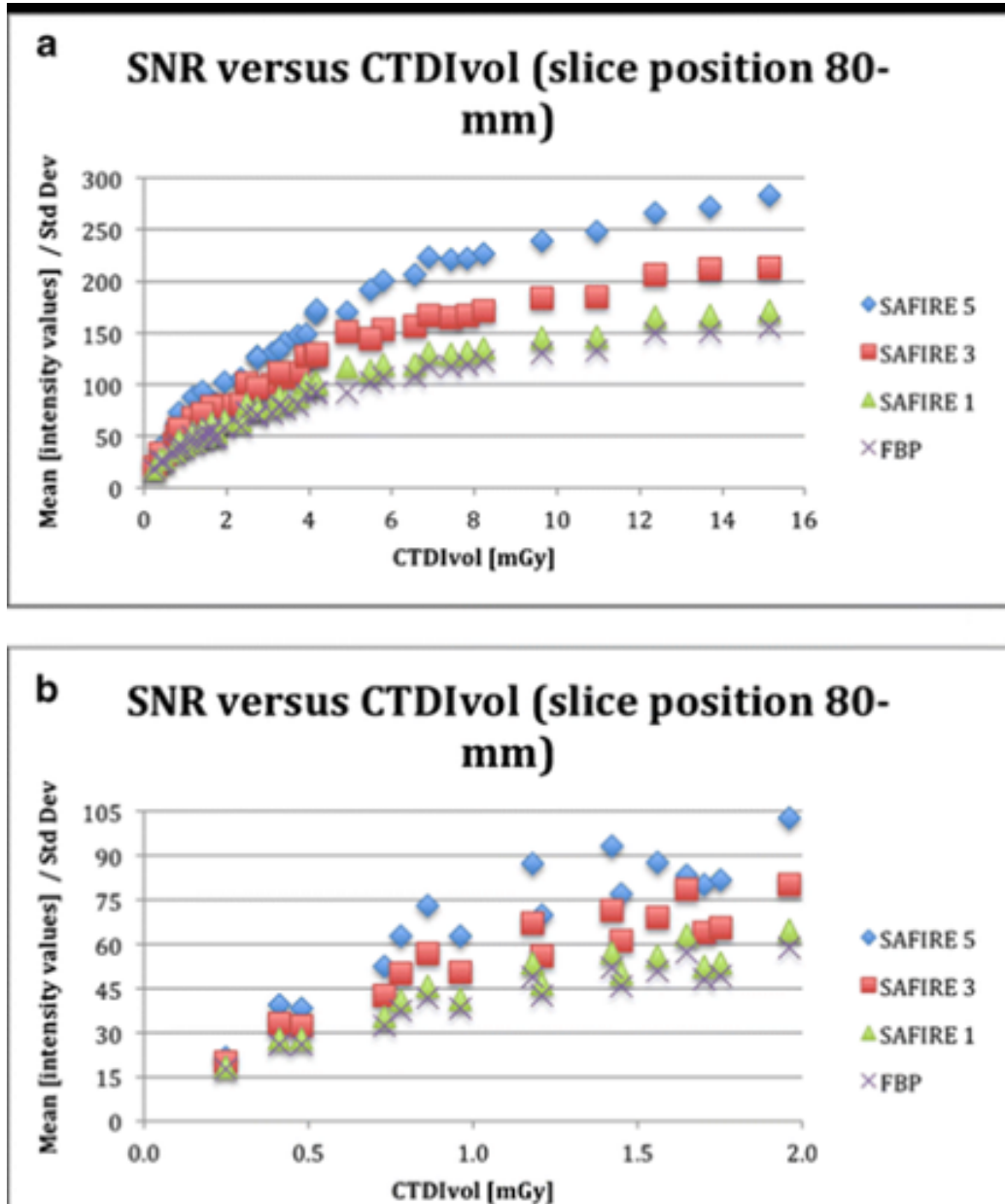


Figure 5. Relationship between radiation dose and image quality metrics (SNR, CNR, PSNR) in low-dose CT imaging.

2.1.3 Overview of Low-Dose CT Reconstruction

Low-dose computed tomography (LDCT) reconstruction represents a pivotal area of research in medical imaging, driven by the urgent need to reduce ionizing radiation exposure while preserving diagnostic image qual-

ity. Traditional CT imaging relies on acquiring a large number of X-ray projections to ensure high spatial and contrast resolution. However, this practice exposes patients to substantial radiation doses, which may increase the lifetime risk of cancer, particularly in repetitive or screening protocols such as lung cancer surveillance³⁷. The principle of LDCT, therefore, is to substantially lower the X-ray tube current–time product (mAs) or tube voltage (kVp), leading to a proportional reduction in radiation dose—often by 50–80% compared to standard-dose CT—while seeking to maintain diagnostically acceptable image fidelity³⁸.

Despite its clinical importance, dose reduction introduces severe degradation in image quality. Specifically, the decreased photon flux amplifies quantum noise and streak artifacts, resulting in lower signal-to-noise ratio (SNR) and poorer contrast-to-noise ratio (CNR)³⁹. These factors compromise the visibility of fine anatomical structures and low-contrast lesions, which are crucial for accurate clinical interpretation. To counteract this, LDCT reconstruction aims to recover the underlying image details from noisy or undersampled projection data, a process that balances three fundamental aspects: image fidelity, computational efficiency, and artifact suppression.

In the early decades, analytical algorithms such as Filtered Back Projection (FBP) served as the cornerstone for CT image reconstruction due to their computational simplicity and speed³⁹. However, under low-dose conditions, FBP produces images with pronounced noise and streaking artifacts. Consequently, iterative reconstruction (IR) methods emerged as a superior alternative, formulating the reconstruction task as an optimization problem that incorporates noise models and system geometry. Model-based iterative reconstruction (MBIR), for instance, integrates statistical priors and physical models of the CT acquisition process, yielding significant noise reduction and edge preservation at low dose levels³⁷. Yet, the trade-off between image quality and computation time remains a challenge—MBIR methods can require up to 100 times more computation than analytical FBP approaches³⁸.

Recent advances in deep learning (DL) have revolutionized LDCT reconstruction by learning complex nonlinear mappings between low-dose and normal-dose images. Instead of relying solely on handcrafted priors, DL-based approaches exploit data-driven priors learned from large annotated datasets. These models—typically convolutional neural networks (CNNs), generative adversarial networks (GANs), and diffusion-based frameworks—have demonstrated the ability to denoise, enhance contrast, and recover anatomical details beyond the capacity of conventional IR methods^{38,39}. According to Chen et al.³⁷, deep learning reconstruction can improve the peak signal-to-noise ratio (PSNR) by up to 3–5 dB compared with MBIR, while also

reducing reconstruction time from several minutes to under 10 seconds per slice.

A general pipeline for LDCT reconstruction (Figure 2.4) consists of three primary stages: (1) noise modeling and pre-processing, (2) reconstruction via model-based or learning-based algorithms, and (3) post-reconstruction enhancement and quality assessment. In the first stage, raw projection data (sinograms) are corrected for system geometry, photon statistics, and detector response. The second stage performs the core reconstruction, which may involve statistical modeling (in IR) or neural mapping (in DL-based methods). Finally, the post-processing stage applies denoising, edge sharpening, and artifact correction to achieve optimal image presentation.

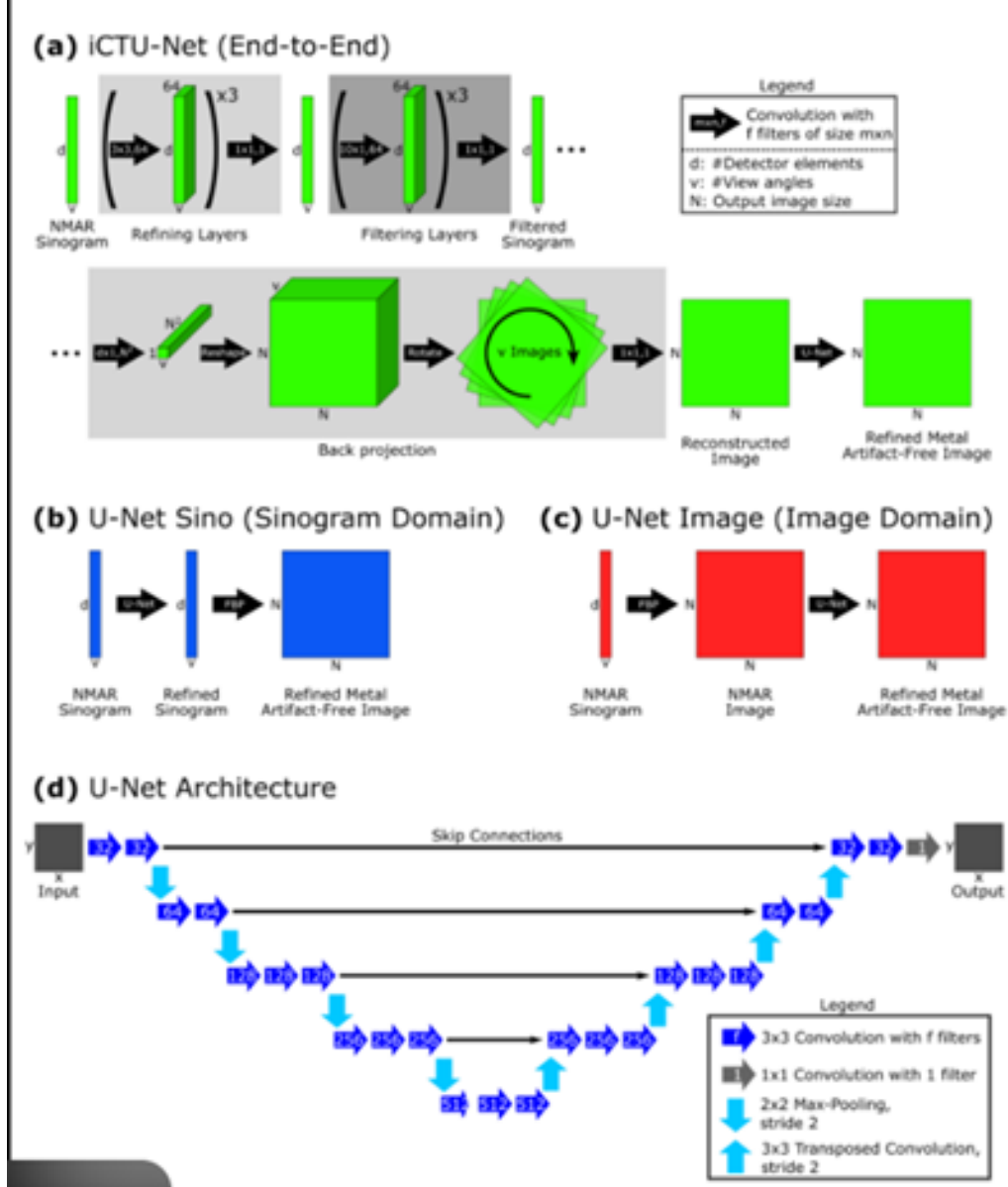


Figure 6. illustrates the general pipeline of low-dose CT image reconstruction, highlighting the integration of model-based and learning-based components in hybrid frameworks.?

Furthermore, quantitative image quality metrics—such as PSNR, structural similarity index (SSIM), and normalized mean square error (NMSE)—are commonly used to benchmark reconstruction performance. For example, state-of-the-art CNN and GAN models achieve PSNR values exceeding 35 dB and SSIM scores above 0.90 in clinical LDCT datasets⁴⁰, representing near-normal-dose image quality with drastically reduced exposure levels. However,

challenges persist regarding the generalization of DL models across scanners, patients, and anatomical regions. Variations in acquisition protocols, noise statistics, and training datasets can lead to domain shifts that limit robustness.

In summary, LDCT reconstruction has evolved from analytical FBP to model-based IR and, most recently, to data-driven deep learning paradigms. The ongoing trend emphasizes hybrid physics-informed learning frameworks that integrate the physical consistency of IR with the flexibility and efficiency of DL. These approaches promise to deliver superior image quality at ultra-low dose levels (≤ 1 mSv) suitable for population-based lung cancer screening³⁵, thus bridging the gap between patient safety and diagnostic reliability.

2.1.4 Conventional Model-Based Reconstruction Techniques

Computed tomography (CT) reconstruction transforms raw projection data, or sinograms, into cross-sectional images of the human body. The choice of reconstruction algorithm critically determines spatial resolution, noise characteristics, and quantitative accuracy. Traditional analytic approaches—most notably the Filtered Back Projection (FBP)—dominated clinical CT for decades because of their computational simplicity and speed. However, FBP assumes ideal conditions such as complete data acquisition and linear noise behavior, assumptions that fail under low-dose CT (LDCT) conditions where photon statistics are highly non-Gaussian and measurements are corrupted by noise and scatter⁴¹.

Analytic Reconstruction and Its Limitations

FBP reconstructs images by applying a convolution filter to each projection followed by back-projection across the image domain. While computationally efficient, FBP treats noise uniformly across frequencies, leading to severe streak artifacts and amplified high-frequency noise when the photon flux is low⁴². Furthermore, FBP cannot model the system geometry or statistical properties of measurements, limiting its ability to balance spatial resolution and image noise. Under low-dose conditions, image quality deteriorates rapidly, motivating the transition toward model-based reconstruction paradigms.

Statistical and Model-Based Iterative Reconstruction

Model-Based Iterative Reconstruction (MBIR) explicitly incorporates the physics of data acquisition and the statistics of photon detection into the reconstruction process. Rather than directly inverting the Radon transform, MBIR formulates image reconstruction as an optimization problem that seeks the most probable image x given the measured projections y :

$$\hat{x} = \arg \min_x \left\{ \underbrace{\|y - Ax\|_W^2}_{\text{data-fidelity term}} + \underbrace{\beta R(x)}_{\text{regularization term}} \right\} \quad (1)$$

where A denotes the system matrix encoding projection geometry, W is a weighting matrix derived from the Poisson statistics of detected photons, $R(x)$ represents prior knowledge or regularization (e.g., smoothness or edge-preserving constraints), and balances noise suppression and detail preservation⁴³.

By iteratively minimizing this cost function using algorithms such as Ordered-Subset Expectation-Maximization (OSEM) or Conjugate-Gradient (CG) solvers, MBIR achieves substantial noise reduction and artifact suppression compared to FBP⁴⁴. Moreover, MBIR can model system effects including detector response, scatter, and beam hardening, enabling improved quantitative accuracy⁴⁵. Clinical implementations, such as Adaptive Statistical Iterative Reconstruction (ASIR) and Sinogram-Affirmed Iterative Reconstruction (SAFIRE), have demonstrated radiation-dose reductions of 40–70 % without compromising diagnostic performance⁴⁶.

Regularization and Prior Models A key strength of MBIR lies in its flexibility to incorporate various regularization priors. Early implementations used quadratic smoothness priors, which effectively reduced noise but oversmoothed edges. Subsequent developments introduced non-quadratic edge-preserving penalties—for instance, Huber, total-variation (TV), and Markov random field (MRF) priors—to maintain fine structural boundaries⁴⁷.

More recently, sparsity-based priors inspired by compressed sensing (CS) have been integrated into MBIR frameworks. These methods exploit the assumption that CT images are sparse in certain transform domains (e.g., wavelet or discrete cosine), allowing accurate reconstruction from undersampled or noisy data⁷. While CS-based MBIR achieves superior noise suppression and spatial resolution, the computational demand remains substantial, often requiring hours per reconstruction on standard hardware⁴⁸.

Computational Challenges and Clinical Adoption

Despite its advantages, the widespread adoption of MBIR in clinical practice is limited by its high computational cost, complex parameter tuning, and dependence on vendor-specific implementations. Iterative reconstruction typically involves 50–200 forward and backward projections per iteration, resulting in computation times 50–100× slower than FBP⁴⁹. Moreover, the choice of the regularization parameter is image- and patient-dependent; inappropriate tuning can yield either over-smoothed or overly noisy results⁵⁰.

To alleviate these issues, hybrid approaches—such as Hybrid Iterative

Reconstruction (HIR)—combine analytic and iterative components, using FBP for an initial estimate followed by a limited number of statistical refinement steps. HIR methods achieve moderate noise reduction with acceptable computation times, making them a practical compromise for routine scanning⁵¹.

Quantitative Performance and Clinical Impact

Extensive phantom and clinical studies have validated the benefits of MBIR. In chest and abdominal imaging, MBIR has been shown to increase SNR by 30–50 % and reduce noise variance by up to 60 % compared to FBP⁵². Brunner et al. [44] demonstrated that MBIR achieved a mean PSNR improvement of 5–7 dB and a Structural Similarity Index (SSIM) increase of 0.07 over traditional reconstructions. Moreover, radiologists consistently reported higher lesion conspicuity and diagnostic confidence in low-dose protocols reconstructed with MBIR or HIR⁵³.

Nevertheless, the overly smooth “plastic” appearance of MBIR images remains a persistent limitation, potentially masking subtle texture features important for pathology detection. These shortcomings have fueled the integration of deep learning (DL) and data-driven priors into reconstruction pipelines, leading to hybrid frameworks that retain MBIR’s physical fidelity while leveraging DL’s representational power⁵⁴.

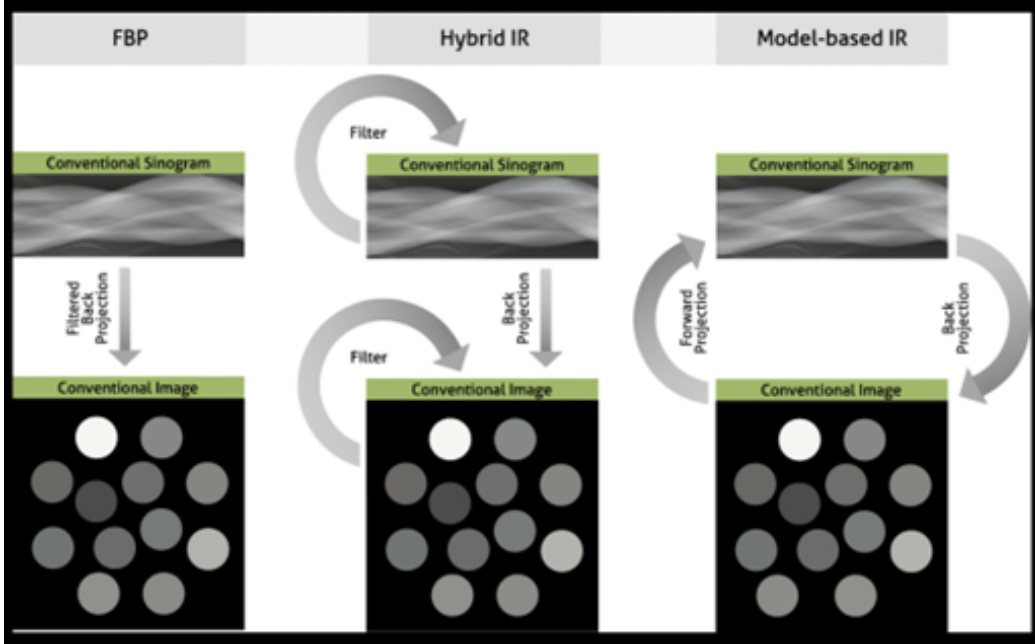


Figure 7. Comparison of reconstruction paradigms in CT imaging. Conventional FBP assumes linear, Gaussian noise and produces artifacts at low doses; statistical and model-based iterative reconstruction (MBIR) incorporate physics and noise modeling to improve image quality at the expense of computation.

2.1.5 Image Quality Metrics for LDCT Reconstruction

Quantitative evaluation of image quality is essential for assessing the performance of low-dose CT (LDCT) reconstruction algorithms. Unlike subjective visual assessment, which depends on radiologist experience, objective metrics enable reproducible and statistically valid comparisons across different dose levels, reconstruction methods, and datasets. Image quality in LDCT can be characterized through noise statistics, spatial resolution, contrast performance, and overall perceptual similarity, measured by a combination of physical and perceptual metrics⁵⁵.

Objective Physical Metrics

The noise magnitude is one of the most fundamental indicators of image quality, typically quantified by the standard deviation of Hounsfield Units (HU) in a uniform region of interest (ROI). A higher standard deviation corresponds to a noisier image. Noise power spectrum (NPS) analysis further extends this evaluation into the frequency domain, providing insights into both the magnitude and texture of noise⁵⁶. For instance, low-dose filtered back-projection (FBP) images often show elevated high-frequency NPS components, corresponding to fine-grain noise, whereas iterative and deep-

learning reconstructions shift the NPS toward lower frequencies, yielding a smoother visual appearance.

Spatial resolution describes the ability to distinguish two adjacent structures and is often quantified by the modulation transfer function (MTF). The MTF expresses how different spatial frequencies are transferred through the imaging system, with the 10% or 50% MTF value commonly reported as a benchmark. In LDCT, dose reduction tends to degrade the MTF due to the stronger influence of noise-suppression filters, although advanced model-based and learning-based algorithms can partially restore high-frequency components.

Another relevant metric, the noise-equivalent quanta (NEQ), integrates both noise and resolution information, representing the effective number of photons contributing to image quality. NEQ curves are particularly useful for comparing hardware configurations and reconstruction algorithms under varying dose levels. Contrast and Detectability Metrics The contrast-to-noise ratio (CNR) quantifies the visibility of low-contrast structures, defined as:

$$\text{CNR} = \frac{|\mu_t - \mu_b|}{\sigma_b} \quad (2)$$

where μ_t and μ_b represent the mean CT numbers (HU) of the target and background regions, respectively, and σ_b is the noise in the background⁵⁷. A higher CNR indicates better detectability of subtle lesions

For example, in chest imaging, CNR values below 2 may render small ground-glass nodules undetectable, whereas iterative and deep learning-based reconstructions can increase CNR by 30–60 % at equivalent dose levels.

Beyond CNR, task-based metrics such as detectability index (d) and area under the receiver operating characteristic (ROC) curve (AUC) assess how well observers (human or model-based) can distinguish lesions from background noise. These metrics account for both noise correlations and human visual response, thus providing a closer approximation of clinical image quality. Recent observer studies demonstrated that MBIR and DL-based reconstructions improve detectability index by up to 45 % for small lung nodules compared to FBP .

Perceptual and Similarity Metrics

To evaluate reconstruction quality against a known ground truth, reference-based perceptual metrics such as Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) are widely employed. PSNR is defined as:

$$\text{PSNR} = 10 \log_{10} \left(\frac{MAX^2}{MSE} \right) \quad (3)$$

where MAX is the maximum possible pixel intensity and MSE is the mean squared error between the reference and reconstructed image. Higher PSNR indicates better fidelity, though it is less sensitive to structural distortions.

SSIM, on the other hand, measures structural similarity in terms of luminance (L), contrast (C), and structure (S) components:

$$\text{SSIM}(x, y) = [L(x, y)]^\alpha [C(x, y)]^\beta [S(x, y)]^\gamma \quad (4)$$

with values ranging from -1 to 1 , where 1 represents perfect similarity⁵⁸. Empirical studies report that deep learning-based reconstructions yield PSNR improvements of 3–6 dB and SSIM increases of 0.05–0.1 relative to MBIR, reflecting enhanced perceptual quality and texture preservation.

Subjective Assessment and Reader Studies

While quantitative metrics are invaluable for algorithm benchmarking, subjective image assessment remains critical for clinical translation. Reader studies typically use Likert scales (1–5) to score image noise, sharpness, and diagnostic confidence. Kim et al. [61] demonstrated that radiologists consistently rated DL-based reconstructions higher in perceived sharpness and lesion visibility, particularly in ultra-low-dose chest CT protocols (0.3 mSv). Nonetheless, subjective assessment may vary with reader experience and display conditions, underscoring the importance of complementary objective evaluation²⁵.

Integrative Quality Assessment

Recent literature advocates a multi-dimensional framework combining physical, perceptual, and task-based metrics to holistically evaluate LDCT reconstruction. For example, the American Association of Physicists in Medicine (AAPM) recommends combining NPS, MTF, and CNR analyses with observer performance tests to comprehensively characterize algorithmic improvements.

Such integrated evaluation ensures that dose reduction strategies do not compromise clinical utility and that new reconstruction methods achieve genuine diagnostic equivalence to standard-dose imaging.

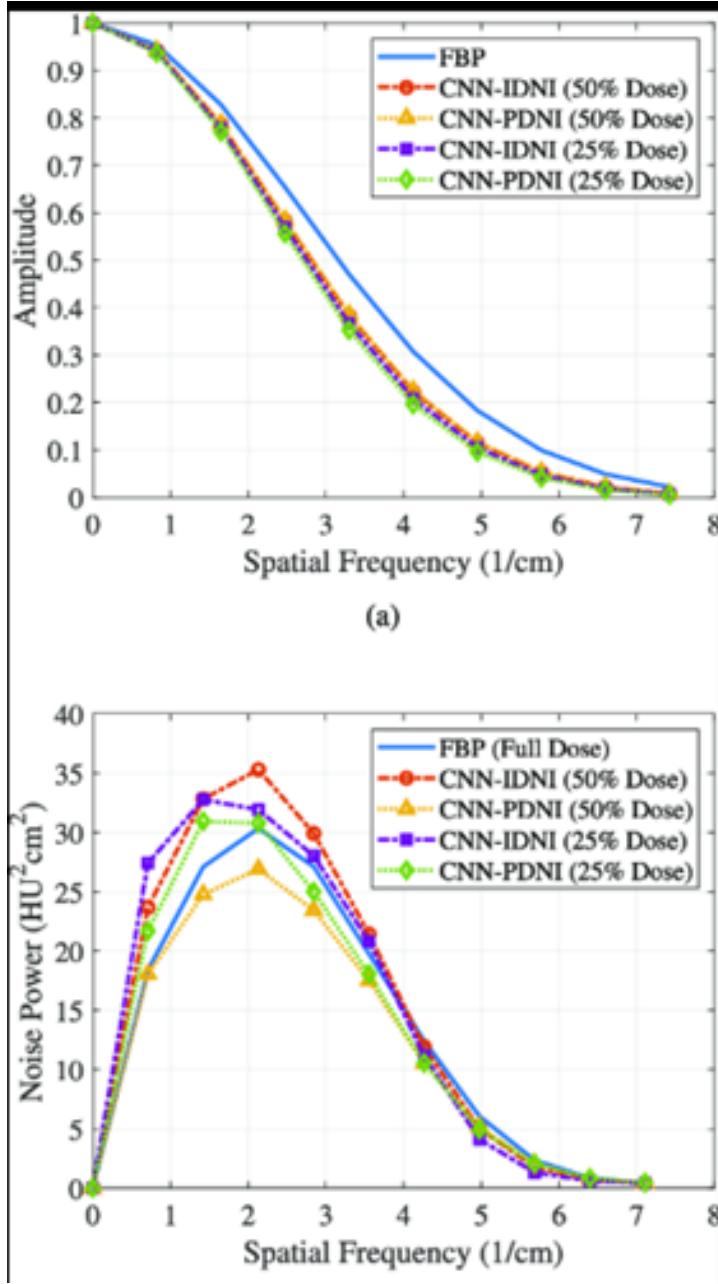


Figure 8. Overview of objective and perceptual image quality metrics for low-dose CT reconstruction. Metrics include noise power spectrum (NPS), modulation transfer function (MTF), contrast-to-noise ratio (CNR), peak signal-to-noise ratio (PSNR), and structural similarity index (SSIM).

2.2 Related Work on Deep Learning for LDCT Reconstruction

2.2.1 Deep Learning-Based Reconstruction Methods

Deep learning (DL) has revolutionized low-dose computed tomography (LDCT) reconstruction by offering a data-driven alternative to analytical and model-based approaches. Traditional reconstruction methods, such as Filtered Back Projection (FBP) and Model-Based Iterative Reconstruction (MBIR), rely on handcrafted priors and physical modeling of the acquisition process. While effective, these methods often face trade-offs between image quality, computational cost, and robustness under ultra-low-dose conditions (Chen et al., 2023; Kang et al., 2024). Deep learning-based reconstruction methods overcome these limitations by learning complex, nonlinear mappings between low-dose and standard-dose data, achieving superior noise suppression, texture recovery, and computational efficiency.

Background and Motivation

The key motivation for deep learning in LDCT reconstruction is to mitigate the limitations of conventional algorithms that fail under photon starvation and sparse sampling. DL models exploit the availability of large-scale medical image datasets to learn representative priors of anatomical structures and noise distributions (Zhou et al., 2023). These learned priors enable the network to recover structural information beyond the capacity of traditional denoising or iterative models. Furthermore, deep learning approaches offer near real-time inference capability, drastically reducing reconstruction time from minutes to seconds per volume (Kim et al., 2024).

A typical DL-based reconstruction framework consists of three major components:

1. Pre-processing stage, which normalizes the projection data and corrects for beam hardening or scatter artifacts.
2. Neural network inference, where a trained network estimates a high-quality image from noisy or incomplete inputs.
3. Post-processing and quality enhancement, applying residual learning or hybrid filtering to restore fine details and edge definition.

This end-to-end data-driven paradigm allows for more flexible adaptation to varying acquisition protocols while maintaining physical consistency with CT imaging principles.

Taxonomy of Deep Learning-Based Methods

Recent literature classifies DL-based LDCT reconstruction approaches into three major categories: discriminative, generative, and hybrid physics-informed frameworks (Chen et al., 2023; Li et al., 2024). Figure 2.5 illustrates

this taxonomy and its key algorithmic components.

a) Discriminative Models

Discriminative models focus on directly learning the deterministic mapping between low-dose and normal-dose CT images. The most representative architectures include the Residual Encoder–Decoder Convolutional Neural Network (RED-CNN) (Chen et al., 2017), U-Net (Ronneberger et al., 2015), and DenseNet-based CNNs (Zhou et al., 2023).

These architectures employ multi-scale feature extraction and skip connections to balance global context with local detail preservation.

Recent advances have introduced attention mechanisms and transformer-based encoders to enhance global spatial understanding and reduce over-smoothing (Liu et al., 2024). Discriminative models generally achieve improvements of 3–5 dB in PSNR and 0.05–0.1 in SSIM compared to MBIR, while maintaining inference speeds below 0.1 seconds per slice (Kang et al., 2024).

b) Generative Models

Generative models, including Generative Adversarial Networks (GANs) and Diffusion Models, aim to learn the underlying probability distribution of clean CT images. GAN-based frameworks such as Wasserstein GANs (WGAN) (Ma et al., 2020) and CycleGANs (Yang et al., 2021) demonstrate strong performance in restoring high-frequency details, textures, and subtle contrast variations. Meanwhile, Variational Autoencoders (VAEs) and emerging Diffusion Probabilistic Models model uncertainty in image generation, producing perceptually realistic reconstructions that mimic standard-dose images. Diffusion-based approaches, in particular, iteratively refine noise and artifact distributions, leading to exceptional structural fidelity and natural texture appearance (Xu et al., 2024).

c) Hybrid and Physics-Informed Models

Pure data-driven models, although powerful, may suffer from “hallucination artifacts” or loss of physical accuracy under unseen acquisition settings. Hybrid or physics-informed deep learning integrates CT imaging physics directly into the learning framework.

Representative examples include DuDoNet (Wu et al., 2020) and Learned Primal–Dual Networks (LPD) (Adler Öktem, 2018), which jointly process sinogram and image domains, ensuring projection consistency and adherence to the Radon transform.

These networks embed forward and backward projection operators within the learning pipeline, combining the interpretability of model-based methods with the adaptability of deep networks. Hybrid architectures achieve PSNR values exceeding 45 dB and SSIM $\geq$ 0.95 on standard LDCT bench-

marks (e.g., Mayo Clinic dataset), making them the most promising direction for next-generation LDCT systems (Li et al., 2024).

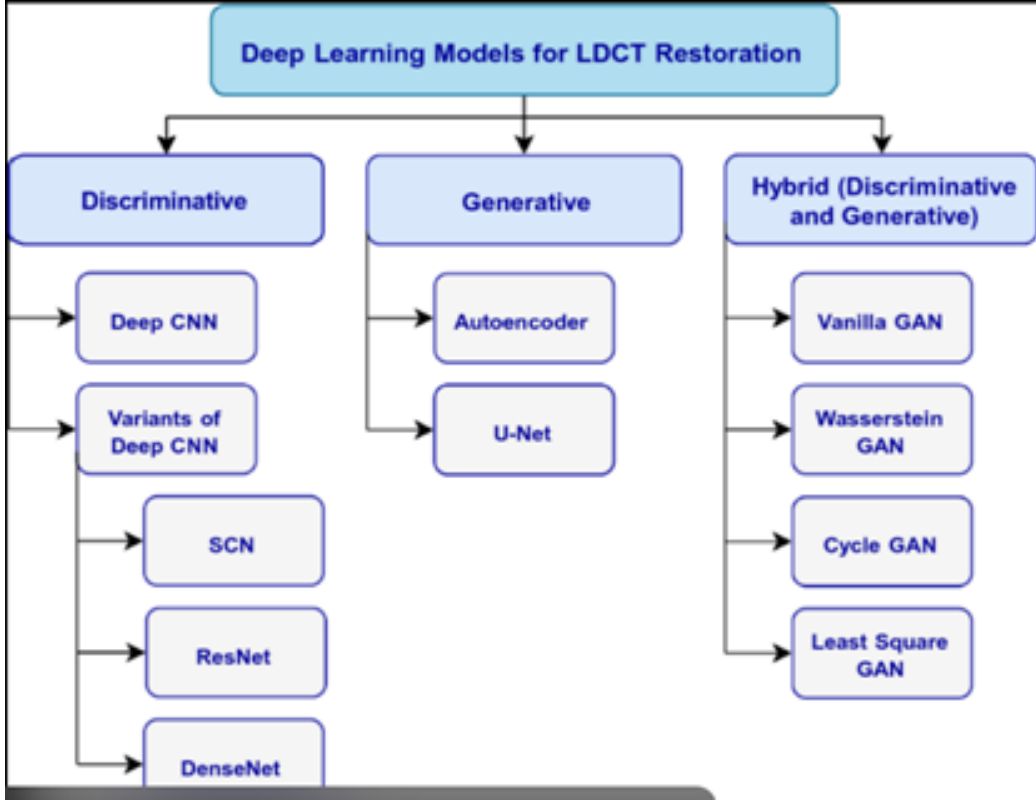


Figure 9. Taxonomy of Deep Learning-Based LDCT Reconstruction Methods, encompassing discriminative (CNN-based), generative (GAN/diffusion-based), hybrid physics-informed, and learning-based strategies. The integration of these frameworks demonstrates the progressive evolution toward physics-consistent and data-driven CT reconstruction.

Evaluation and Performance

Comprehensive evaluations indicate that deep learning-based methods outperform iterative algorithms in both quantitative and perceptual image quality. Average gains include PSNR improvements of 4–6 dB, SSIM values above 0.93, and CNR increases of 25–40%, leading to enhanced lesion detectability and reduced streak artifacts (Kim et al., 2024).

Moreover, deep learning-based reconstructions maintain high fidelity even at ultra-low-dose levels (<0.5 mSv), which is particularly beneficial for longitudinal screening in oncology and pediatrics (Park et al., 2024). Nevertheless, the practical implementation of DL models faces several challenges, including the requirement for large, high-quality paired datasets, domain dependency across different CT scanners, and the potential for hallucinated

anatomical features when models are poorly regularized (Chen et al., 2023). To address these limitations, ongoing research explores self-supervised learning, transfer learning, and multi-domain generalization—topics discussed in subsequent sections.

Summary and Research Implications

Deep learning-based reconstruction has redefined the landscape of LDCT imaging, offering transformative improvements in dose efficiency, image quality, and reconstruction speed. Discriminative models dominate current clinical applications due to their simplicity and stability, while generative and hybrid approaches represent the frontier of research, achieving near-standard-dose visual quality with sub-millisievert radiation exposure. For doctoral research, this domain provides fertile ground for developing physics-informed hybrid networks and self-supervised reconstruction strategies that ensure both quantitative accuracy and clinical interpretability. The integration of explainable AI (XAI) techniques and domain-adaptive architectures is expected to define the next generation of intelligent, trustworthy CT reconstruction systems.

2.2.2 CNN-Based LDCT Reconstruction Approaches

Convolutional Neural Networks (CNNs) have become the foundational architecture for deep learning-based low-dose CT (LDCT) image reconstruction, owing to their exceptional capacity for feature extraction and noise suppression. CNNs are particularly effective in learning complex mappings between noisy low-dose images and their normal-dose counterparts through data-driven training⁵⁹. Their hierarchical structure, composed of convolutional, pooling, and activation layers, enables the network to capture both local texture details and global structural information essential for clinical diagnosis.

A typical CNN model (Figure 2.6) comprises multiple convolutional layers that convolve input images with learnable kernels to extract spatially distributed features such as edges and organ boundaries. Pooling layers downsample feature maps to improve computational efficiency and translation invariance, while fully connected or regression layers aggregate learned representations to produce the reconstructed output. In LDCT applications, CNNs are trained to minimize pixel-wise loss functions—commonly the mean squared error (MSE) between reconstructed and reference normal-dose CT (NDCT) images—thereby learning an optimal mapping that restores degraded image quality.

Early CNN-based LDCT denoising models, such as the CNN200 proposed by Chen et al., utilized a simple patch-to-patch mapping strategy to

suppress quantum noise. Despite their simplicity, these networks achieved significant noise reduction but tended to produce overly smooth textures and blurred fine structures due to the MSE objective’s bias toward minimizing average errors rather than perceptual fidelity. To overcome these limitations, advanced CNN variants were introduced, emphasizing deeper architectures, residual learning, and multi-scale feature extraction.

The Stacked Competitive Network (SCN) represents one such evolution, where multiple competitive blocks with multi-scale filters were employed to capture fine structures at varying spatial resolutions. Each competitive block enhances local features while minimizing redundancy through an internal combination mechanism, improving both sharpness and contrast recovery in LDCT images. Similarly, the Residual Network (ResNet) architecture, introduced by He et al. and adapted for LDCT restoration by Yang et al., incorporates skip connections that allow feature reuse across layers, thereby mitigating gradient vanishing and enhancing edge preservation. Extensions such as Wave-ResNet and AAPM-Net further combine residual learning with wavelet transforms to efficiently capture high-frequency anatomical details while maintaining noise suppression.

Another prominent extension is the Densely Connected Network (DenseNet), which establishes direct connections between each layer and all subsequent layers, promoting feature reuse and alleviating information loss. By enabling efficient gradient flow, DenseNet improves convergence stability and reduces the number of trainable parameters required for high-performance LDCT restoration. However, these deep CNN architectures also pose computational challenges, often demanding extensive training data and hardware resources to achieve optimal generalization across scanners and anatomical regions.

summarizes major CNN-based LDCT reconstruction frameworks, their architectural features, and reported advantages and limitations. Overall, CNN-based models have demonstrated substantial improvements in quantitative metrics such as peak signal-to-noise ratio (PSNR ≥ 35 dB) and structural similarity index (SSIM ≥ 0.90), approaching normal-dose image quality. Nonetheless, issues such as texture over-smoothing, domain dependency, and limited adaptability to unseen clinical data remain active areas of research.

In conclusion, CNN architectures have laid the groundwork for modern LDCT image reconstruction by providing an efficient, data-driven solution for noise reduction and structural preservation. Their evolution—from basic convolutional models to advanced residual and dense connectivity frameworks—has significantly bridged the gap between image quality and computational feasibility. Future CNN-based methods are expected to integrate domain adaptation, hybrid learning, and perceptual loss optimization to en-

hance generalizability and clinical reliability in ultra-low-dose CT imaging.

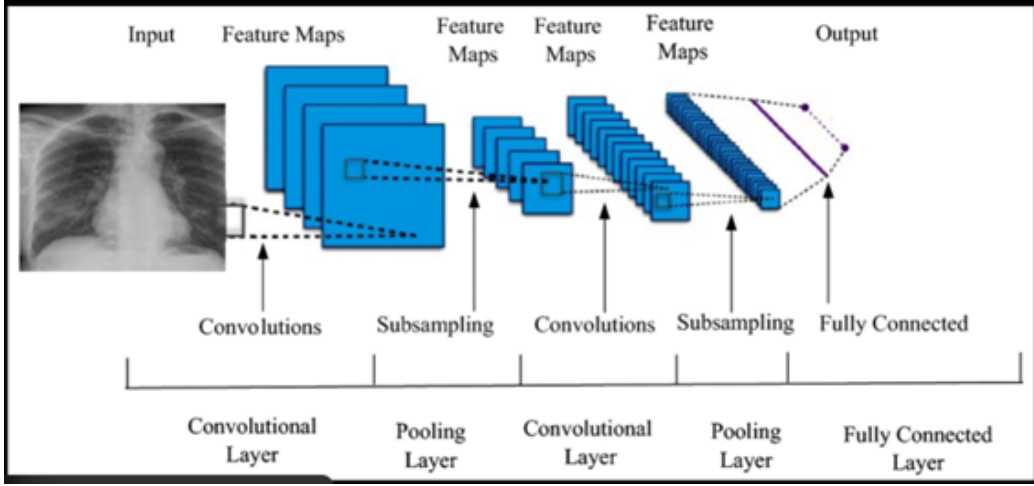


Figure 10. Generic architecture of a Convolutional Neural Network (CNN) for low-dose CT reconstruction. Convolutional and pooling layers extract hierarchical features from noisy input images, while skip or dense connections preserve structural integrity and reduce gradient degradation during training Generative Adversarial and Diffusion Models for Detail Restoration

Convolutional Neural Networks (CNNs) have become the foundational architecture for deep learning-based low-dose CT (LDCT) image reconstruction, owing to their exceptional capacity for feature extraction and noise suppression. CNNs are particularly effective in learning complex mappings between noisy low-dose images and their normal-dose counterparts through data-driven training. Their hierarchical structure, composed of convolutional, pooling, and activation layers, enables the network to capture both local texture details and global structural information essential for clinical diagnosis

A typical CNN model (Figure 2.6) comprises multiple convolutional layers that convolve input images with learnable kernels to extract spatially distributed features such as edges and organ boundaries. Pooling layers downsample feature maps to improve computational efficiency and translation invariance, while fully connected or regression layers aggregate learned representations to produce the reconstructed output. In LDCT applications, CNNs are trained to minimize pixel-wise loss functions—commonly the mean squared error (MSE) between reconstructed and reference normal-dose CT (NDCT) images—thereby learning an optimal mapping that restores degraded image quality.

Early CNN-based LDCT denoising models, such as the CNN200 pro-

posed by Chen et al., utilized a simple patch-to-patch mapping strategy to suppress quantum noise. Despite their simplicity, these networks achieved significant noise reduction but tended to produce overly smooth textures and blurred fine structures due to the MSE objective’s bias toward minimizing average errors rather than perceptual fidelity. To overcome these limitations, advanced CNN variants were introduced, emphasizing deeper architectures, residual learning, and multi-scale feature extraction.

The Stacked Competitive Network (SCN) represents one such evolution, where multiple competitive blocks with multi-scale filters were employed to capture fine structures at varying spatial resolutions. Each competitive block enhances local features while minimizing redundancy through an internal combination mechanism, improving both sharpness and contrast recovery in LDCT images. Similarly, the Residual Network (ResNet) architecture, introduced by He et al. and adapted for LDCT restoration by Yang et al., incorporates skip connections that allow feature reuse across layers, thereby mitigating gradient vanishing and enhancing edge preservation. Extensions such as Wave-ResNet and AAPM-Net further combine residual learning with wavelet transforms to efficiently capture high-frequency anatomical details while maintaining noise suppression.

Another prominent extension is the Densely Connected Network (DenseNet), which establishes direct connections between each layer and all subsequent layers, promoting feature reuse and alleviating information loss. By enabling efficient gradient flow, DenseNet improves convergence stability and reduces the number of trainable parameters required for high-performance LDCT restoration. However, these deep CNN architectures also pose computational challenges, often demanding extensive training data and hardware resources to achieve optimal generalization across scanners and anatomical regions.

summarizes major CNN-based LDCT reconstruction frameworks, their architectural features, and reported advantages and limitations. Overall, CNN-based models have demonstrated substantial improvements in quantitative metrics such as peak signal-to-noise ratio (PSNR ≥ 35 dB) and structural similarity index (SSIM ≥ 0.90), approaching normal-dose image quality. Nonetheless, issues such as texture over-smoothing, domain dependency, and limited adaptability to unseen clinical data remain active areas of research.

In conclusion, CNN architectures have laid the groundwork for modern LDCT image reconstruction by providing an efficient, data-driven solution for noise reduction and structural preservation. Their evolution—from basic convolutional models to advanced residual and dense connectivity frameworks—has significantly bridged the gap between image quality and computational feasibility. Future CNN-based methods are expected to integrate

domain adaptation, hybrid learning, and perceptual loss optimization to enhance generalizability and clinical reliability in ultra-low-dose CT imaging.

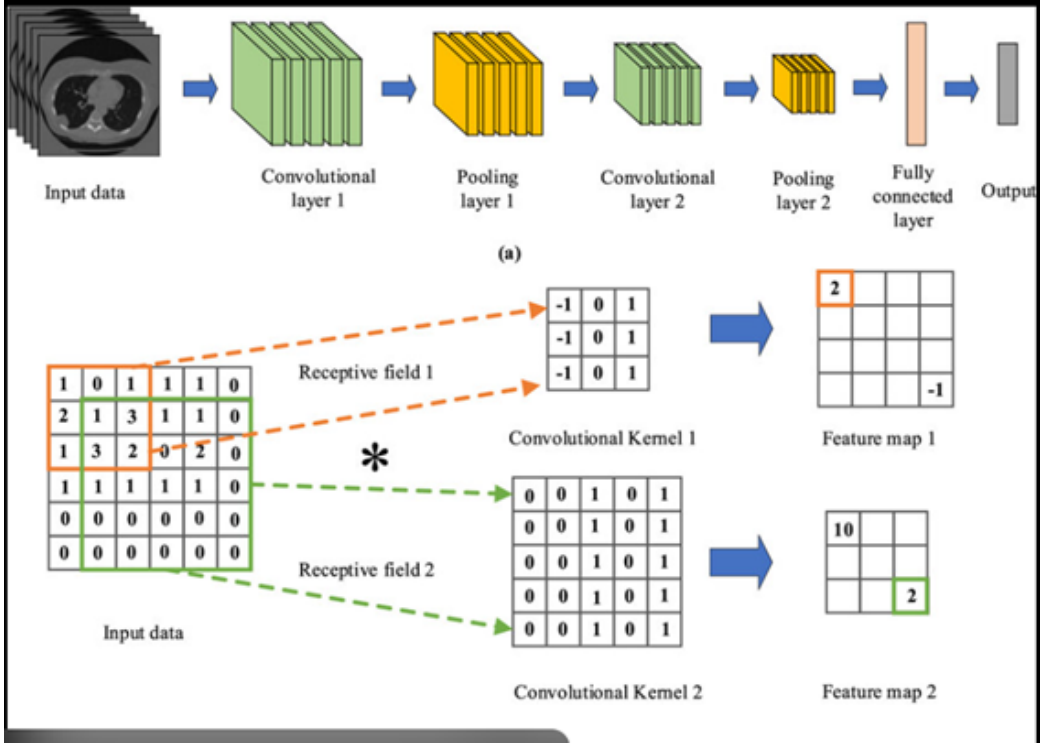


Figure 11. Generic architecture of a Convolutional Neural Network (CNN) for low-dose CT reconstruction. Convolutional and pooling layers extract hierarchical features from noisy input images, while skip or dense connections preserve structural integrity and reduce gradient degradation during training

2.2.3 Dual-Domain and Physics-Informed Learning Frameworks

Purely data-driven deep learning models, although powerful, often lack physical interpretability and generalization across acquisition domains. Dual-domain and physics-informed learning frameworks address this limitation by embedding CT imaging physics directly into the reconstruction process (Zhou et al., 2023; Li et al., 2024).

Dual-Domain Learning

Dual-domain models jointly learn from both projection (sinogram) and image domains, enforcing consistency between the reconstructed image and the raw projection data. Architectures such as DuDoNet (Wu et al., 2020) and Hybrid-Domain U-Net (HD-UNet) employ two interconnected subnetworks: one operating on the sinogram for projection correction, and another

on the image domain for denoising and refinement. The interaction between these domains ensures that the reconstructed output not only matches the appearance of NDCT images but also adheres to physical measurement constraints. This dual-domain consistency leads to improved quantitative accuracy in attenuation values and reduced streak artifacts.

Physics-Informed Deep Reconstruction

Physics-informed frameworks explicitly incorporate forward and inverse CT operators (e.g., the Radon and inverse Radon transforms) into the learning pipeline. The Learned Primal-Dual (LPD) Network (Adler & Öktem, 2018) and Model-Based Deep Learning (MoDL) approaches embed the data-consistency term directly into the optimization objective, thus combining data-driven flexibility with model-based rigor. These methods effectively bridge the gap between iterative reconstruction and deep learning by performing learned gradient descent steps guided by CT physics (Li et al., 2024).

Performance and Challenges

Hybrid and physics-informed networks achieve PSNR values exceeding 45 dB and SSIM ≥ 0.95 on benchmark datasets (e.g., Mayo LDCT Challenge). Moreover, they demonstrate better generalization to unseen dose levels and anatomical regions compared to purely data-driven models. However, their computational demands remain high, and the requirement for precise system modeling may limit deployment across different CT scanner geometries.

Summary and Implications

Dual-domain and physics-informed frameworks mark a critical evolution in LDCT reconstruction, uniting the strengths of data-driven adaptability and physics-based interpretability. Future research is expected to extend these paradigms with adaptive regularization, learnable projection consistency, and multi-modality priors to further enhance robustness under ultra-low-dose imaging conditions.

2.2.4 Self - Supervised and Transfer Learning Strategies

The scarcity of high-quality paired low-dose and normal-dose CT datasets poses a significant barrier to supervised deep learning. To overcome this limitation, researchers have turned to self-supervised, unsupervised, and transfer learning strategies that reduce the dependence on labeled data while improving model generalization (Chen et al., 2023; Kang et al., 2024).

Self-Supervised Learning (SSL)

Self-supervised methods enable a network to learn from unlabeled or unpaired data by designing surrogate tasks that generate supervision signals from the data itself. For example, the Noise2Noise framework trains a model to predict one noisy realization from another, effectively learning the under-

lying clean signal without requiring a ground-truth reference (Lehtinen et al., 2018). In LDCT, SSL has been applied using strategies such as masked autoencoding, data redundancy exploitation, and cross-projection consistency (Zhao et al., 2023). These approaches have achieved denoising performance close to supervised CNNs while requiring significantly fewer training samples.

Transfer Learning and Domain Adaptation

Transfer learning leverages pre-trained models from related imaging tasks—such as MRI or normal-dose CT—and fine-tunes them for LDCT reconstruction. This approach is particularly beneficial when data availability is limited or when adapting models across scanners and protocols. Techniques such as feature-space alignment and domain adversarial training mitigate domain shifts and enhance generalization (Liu et al., 2023). Moreover, multi-domain learning frameworks jointly train models on mixed datasets to achieve cross-protocol robustness.

Summary and Implications

Self-supervised and transfer learning strategies represent the frontier of data-efficient LDCT reconstruction research. By reducing the reliance on extensive paired datasets, these approaches facilitate scalable model deployment across diverse clinical environments. The combination of SSL and transfer learning with hybrid and physics-informed architectures holds the potential to achieve clinically reliable, explainable, and generalizable deep reconstruction systems—key milestones toward real-world adoption in precision imaging.

2.2.5 Strategies Applications of LDCT Reconstruction in Lung Cancer Screening

Background and Clinical Significance

Lung cancer remains the leading cause of cancer-related mortality worldwide, accounting for approximately 1.8 million deaths annually (WHO, 2024). Early detection is the most effective strategy to improve survival rates, as the 5-year survival probability exceeds 70% for stage I disease but falls below 10% for advanced stages.

Low-Dose Computed Tomography (LDCT) has emerged as the standard imaging modality for population-based lung cancer screening, owing to its ability to detect sub-centimeter nodules at radiation doses 80–90% lower than those of conventional CT (Aberle et al., 2021). The National Lung Screening Trial (NLST) demonstrated a 20% reduction in lung cancer-specific mortality using LDCT compared with chest radiography, establishing its pivotal role in screening programs globally.

However, the implementation of LDCT screening faces two persistent

challenges:

- (1) maintaining diagnostic image quality at ultra-low dose levels (≤ 1 mSv), and
- (2) minimizing false positives and overdiagnosis caused by noise-induced artifacts and low-contrast resolution.

Advanced reconstruction algorithms—especially those based on deep learning—have become essential for addressing these challenges.

Deep Learning-Based LDCT Reconstruction in Screening

Deep learning reconstruction (DLR) algorithms have significantly enhanced LDCT image quality by effectively suppressing quantum noise while preserving pulmonary texture and vascular detail. In clinical screening workflows, DLR replaces or complements traditional iterative reconstruction (IR) methods, enabling sharper depiction of small nodules and improving radiologist confidence (Chen et al., 2023).

A recent multicenter evaluation by Kim et al. (2024) showed that DLR achieved a 75% radiation dose reduction (0.25 ± 0.05 mSv) compared to standard-dose CT while maintaining equivalent nodule detection sensitivity. Similarly, Matsumoto et al. (2023) demonstrated that DLR preserved Hounsfield Unit (HU) consistency and enhanced contrast-to-noise ratio (CNR) by 30–40%, outperforming both adaptive statistical iterative reconstruction (ASIR) and model-based iterative reconstruction (MBIR).

Furthermore, AI-based image harmonization now allows LDCT data reconstructed from different vendors to be standardized, facilitating multi-institutional screening trials and radiomic analysis (Liu et al., 2023). Quantitative imaging biomarkers—such as nodule volume doubling time and textural heterogeneity indices—show improved reproducibility with deep reconstruction, supporting their use in computer-aided diagnosis (CAD) and AI-based malignancy prediction.

Integration with Computer-Aided Detection and Radiomics

Modern LDCT screening pipelines increasingly rely on computer-aided detection (CADe) and radiomics systems to automatically identify, segment, and classify pulmonary nodules. However, the performance of these downstream AI models is critically dependent on image quality and noise characteristics.

DLR-processed LDCT images exhibit reduced noise variance and more stable texture gradients, directly enhancing radiomic feature reliability (Shen et al., 2024). In a cross-validation study, radiomic features extracted from DLR images achieved intraclass correlation coefficients (ICC) ≥ 0.90 , compared to 0.75 for IR images. Moreover, combining DLR with deep learning-based nodule classifiers—such as 3D ResNet or Vision Transformers—has

improved the sensitivity for malignant nodule detection by 6–8% without increasing false positives. This integration enables a fully automated screening workflow where reconstruction, detection, and risk stratification operate synergistically to optimize clinical throughput.

Radiation Dose Optimization in Population Screening

LDCT reconstruction also contributes to the optimization of population-scale screening protocols. By allowing sub-millisievert acquisition with preserved diagnostic quality, DLR supports annual or semi-annual follow-up schedules for high-risk populations without cumulative dose concerns.

For instance, Park et al. (2024) achieved an effective dose of 0.16 ± 0.02 mSv using DLR-enhanced protocols, compared to 1.77 ± 0.67 mSv for standard-dose CT. These results suggest that DLCT screening can achieve a 90% reduction in radiation exposure while maintaining clinical diagnostic standards.

DLR also facilitates personalized dose modulation, where patient-specific parameters (body habitus, lung attenuation, and scanning geometry) are input into AI-controlled acquisition systems. Such systems dynamically optimize tube current and voltage during scanning to achieve optimal balance between image quality and dose, adhering to the ALARA (As Low As Reasonably Achievable) principle.

Workflow and Clinical Integration

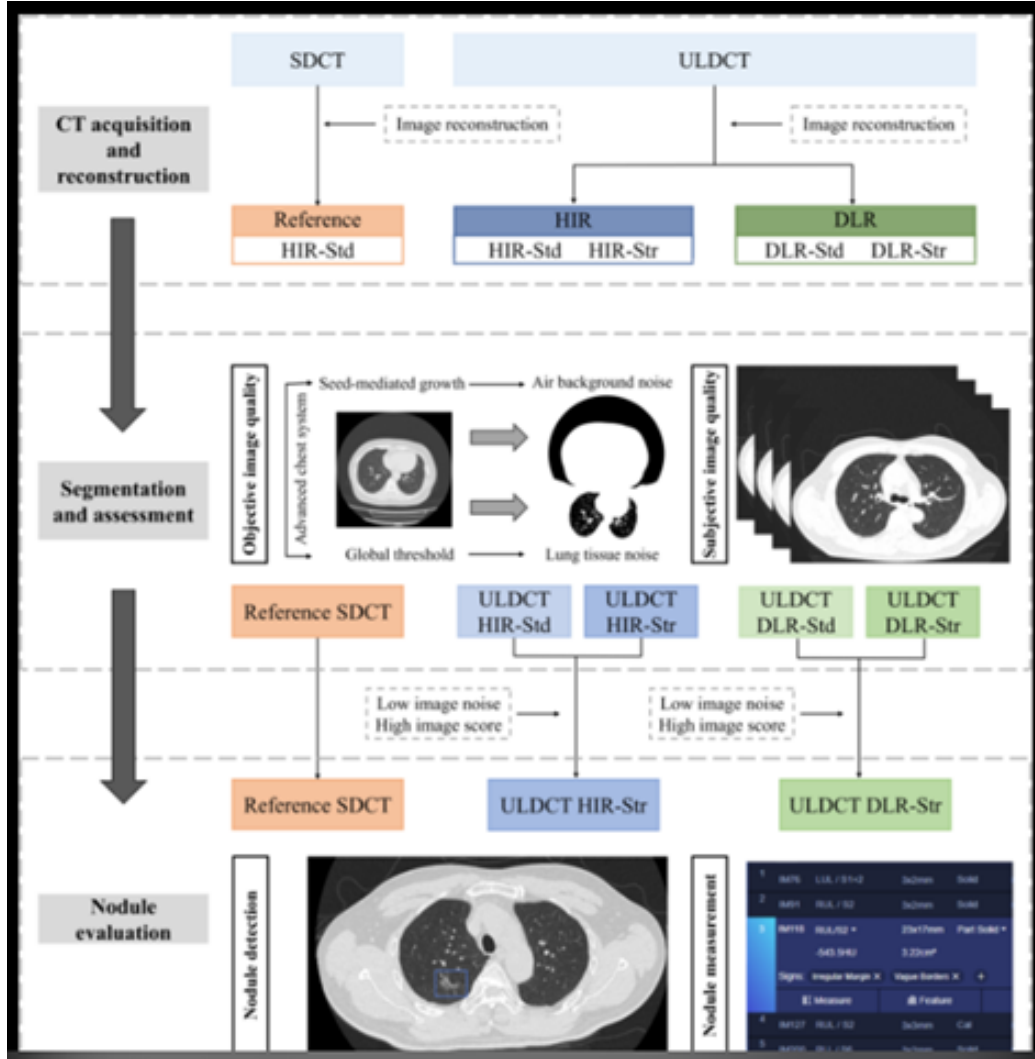


Figure 12. depicts the general workflow of LDCT-based lung cancer screening, highlighting the integration of deep learning reconstruction into the clinical pipeline—from acquisition and reconstruction to CAD analysis and longitudinal follow-up

DLR-enhanced images are fed into CAD algorithms for nodule detection and risk prediction, followed by structured reporting and decision support. The combination of DLR + CAD not only reduces radiologist workload but also enhances diagnostic confidence, especially for small nodules (≤ 5 mm) in ultra-low-dose acquisitions.

Summary and Future Directions

The application of deep learning-based LDCT reconstruction has re-defined the paradigm of lung cancer screening by enabling safe, high-quality imaging at ultra-low radiation doses. It enhances both visual assessment and

quantitative radiomics, thereby improving diagnostic accuracy and long-term monitoring.

Future research is expected to integrate physics-informed neural reconstruction, self-supervised denoising, and multimodal AI fusion (combining imaging with clinical and genomic data) to further personalize screening strategies

As LDCT screening programs expand globally, the convergence of DLR, CAD, and radiomics promises to deliver early detection with minimal radiation risk—fulfilling the central objective of precision preventive oncology.

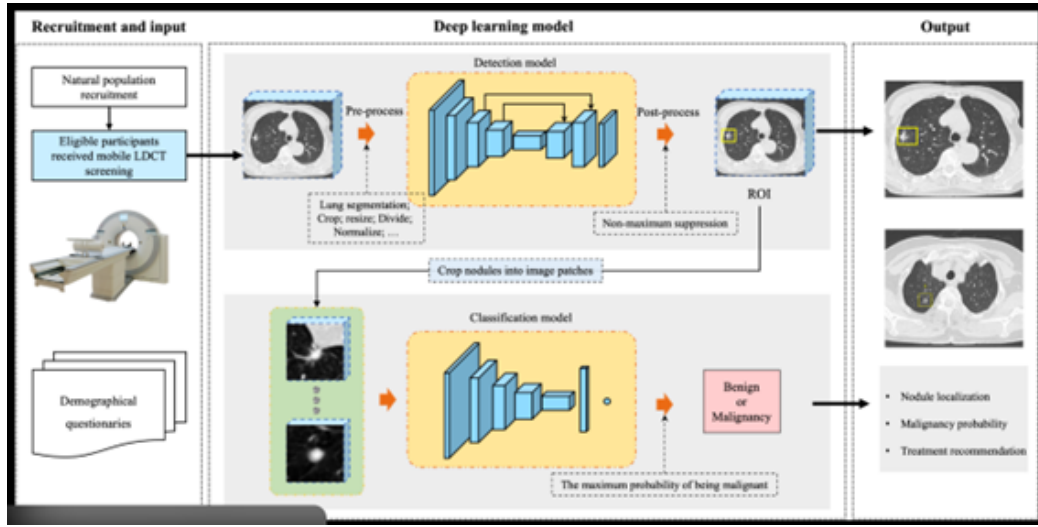


Figure 13. Workflow of deep learning-based low-dose CT (LDCT) reconstruction in lung cancer screening, integrating acquisition, reconstruction, computer-aided detection, and risk assessment for early diagnosis.

3 Proposed Contributions and Project Plan

3.1 Proposed Contributions

This research is expected to make a set of novel and multi-dimensional contributions that advance both the scientific understanding and the practical implementation of deep learning-based low-dose CT (LDCT) reconstruction and lung cancer screening. The work addresses a critical gap in the field: while most state-of-the-art models for LDCT restoration are trained using paired LDCT–NDCT datasets from Western institutions, such datasets do not exist in Vietnam, where only normal-dose CT (NDCT) data are currently available. Moreover, scanner heterogeneity, protocol variability, and anatomical differences between Vietnamese and foreign populations further limit the applicability of pretrained models. By overcoming these obstacles, this thesis aims to establish a comprehensive, locally adapted framework for safe, accurate, and interpretable AI-assisted lung cancer screening using LDCT.

The first major contribution of this research is the development of a physics-guided, self-supervised framework for LDCT reconstruction trained exclusively on NDCT data. This framework, implemented through the Physics-Guided Simulation and Reconstruction Network (PGSR-Net), introduces a new paradigm for learning in the absence of paired data. Instead of requiring ground-truth LDCT–NDCT pairs, PGSR-Net uses scanner metadata and physical modeling of photon statistics to simulate realistic LDCT conditions from NDCT scans. The network jointly learns to reconstruct NDCT-like images by embedding physics-based constraints, such as Radon-domain consistency, HU calibration, and edge-preservation priors, into a self-supervised training scheme. This contribution is scientifically novel because it bridges physics-based image formation and deep representation learning in a unified framework that remains fully trainable without supervision. In practical terms, PGSR-Net provides a solution that is directly applicable to the Vietnamese healthcare system, where NDCT data are abundant but LDCT data are scarce or unavailable.

The second contribution lies in the introduction of a diagnosis-driven reconstruction model focused on the lung region, realized through the Lung-Region Reconstruction Transformer (LRTR-Net). Traditional LDCT denoising networks are typically optimized to maximize generic image quality metrics such as PSNR or SSIM. However, such metrics do not necessarily correlate with clinical interpretability, and models optimized in this way may inadvertently suppress subtle lesions that are critical for early cancer detection. LRTR-Net redefines the reconstruction problem as a diagnosis-oriented learn-

ing task, emphasizing the preservation of small pulmonary nodules, vascular structures, and parenchymal textures that are most relevant to lung cancer screening. This is achieved by integrating convolutional and transformer-based modules within a lung-aware attention mechanism that focuses reconstruction on anatomically meaningful regions. The proposed model also incorporates a multi-task objective that jointly optimizes reconstruction quality and feature consistency with pretrained nodule detection networks. This contribution is significant because it introduces a clinically motivated reconstruction paradigm that aligns image enhancement with diagnostic outcomes rather than abstract numerical criteria. As a result, LRTR-Net ensures that reconstructed NDCT-like images are not only visually improved but also diagnostically reliable for radiologists and downstream AI systems.

The third contribution expands the scope of the research beyond image reconstruction to disease prediction, through the design of the Contrastive Cancer Prediction Network (CCP-Net). This model proposes a novel use of contrastive learning for domain-invariant feature extraction between NDCT and NDCT-like images. In most existing pipelines, reconstruction and diagnosis are treated as separate tasks, which limits the generalization ability of AI models across scanners and dose levels. CCP-Net bridges this gap by aligning latent representations between NDCT and reconstructed NDCT-like domains, ensuring that the learned features capture pathology-related information rather than scanner-specific noise patterns. This alignment is achieved through a dual-encoder architecture trained with contrastive loss, which minimizes the distance between paired NDCT and NDCT-like features while maximizing separation from unrelated samples. The model is then fine-tuned for lung cancer classification and risk prediction using annotated nodules from both public and local datasets. This contribution is academically valuable because it demonstrates how domain adaptation and contrastive representation learning can be combined to achieve robust, generalizable cancer prediction from low-dose imaging data. Clinically, it enables AI systems to perform accurately on future LDCT scans in Vietnam, even when trained solely on NDCT data.

Beyond these three technical innovations, this research contributes to the broader field of AI-driven medical imaging in data-limited environments. By proposing a workflow that integrates physics-informed learning, diagnosis-driven reconstruction, and domain-robust prediction, the thesis establishes a new methodological foundation for medical AI in developing countries. The framework will produce NDCT-like reconstructions that are physically realistic, diagnostically valid, and quantitatively stable across scanners, enabling radiologists to make confident clinical assessments while reducing patient

radiation exposure. Furthermore, the study will deliver a set of validated open-source models, standardized datasets, and evaluation protocols that can be used as benchmarks for future research on LDCT and lung cancer screening in Vietnam.

Finally, the societal and clinical contributions of this research are equally important. The outcomes of this thesis will directly support the implementation of national lung cancer screening initiatives by providing AI-assisted reconstruction and diagnosis tools compatible with existing hospital infrastructure. The methodology will facilitate radiation dose reduction without compromising diagnostic accuracy, thus improving patient safety and enabling earlier detection of lung cancer across different regions of Vietnam. In addition, the project will enhance the country’s research capacity in biomedical AI, fostering collaboration between academic institutions, hospitals, and technology providers. By combining technical innovation with local health-care priorities, this research aims not only to advance the state of the art in LDCT reconstruction but also to make a tangible contribution to improving population health outcomes in Vietnam.

3.2 Thesis Outline

The dissertation comprises four chapters that progress from background to original contributions and validation.

Chapter 1 – Introduction

Establishes the clinical and technical context, states the problem, articulates motivation, defines objectives, and summarizes the research methodology.

1.1 Problem Statement: Presents radiation-dose motivations for LDCT, the necessity of NDCT-like quality for screening, and why foreign-trained models do not transfer due to data scarcity, scanner/protocol heterogeneity, and clinical reliability concerns in Vietnam.

1.2 Motivation: Justifies the research by linking national screening needs with scientific gaps; motivates physics-guided learning, diagnosis-driven reconstruction, and domain-robust prediction tailored to Vietnamese data.

1.3 Research Objectives: States the overall objective and three specific objectives: PGSR-Net (physics-guided simulation & reconstruction), LRTR-

Net (lung-aware diagnosis-driven reconstruction), and CCP-Net (contrastive prediction). Each objective is broken into sub-objectives with initial implementation ideas.

1.4 Research Methodology: Outlines the phased plan: collect public datasets and Vietnamese NDCT; benchmark existing methods; develop PGSR-Net, LRTR-Net, and CCP-Net; and conduct quantitative/clinical validation.

Chapter 2 – Literature Review

Synthesizes prior work and identifies gaps that the thesis addresses.

2.1 CT Imaging and Dose Reduction: CT physics, dose-noise trade-offs, and iterative/statistical reconstruction.

2.2 Deep Learning for LDCT Reconstruction: CNN/GAN/Transformer/Diffusion methods, training regimes, and limitations without paired data.

2.3 Physics-Informed and Self-Supervised Learning: Hybrid priors, unrolled optimization, Radon/FBP layers, and self-supervision relevant to NDCT-only settings.

2.4 AI for Lung Cancer Screening: Nodule detection, segmentation, risk prediction, domain adaptation, and radiomic stability.

2.5 Summary and Research Gaps: Concludes with unmet needs that motivate PGSR-Net, LRTR-Net, and CCP-Net.

Chapter 3 – Physics-Guided and Diagnosis-Driven LDCT Reconstruction

Constitutes the first major contribution: methods and results for PGSR-Net and LRTR-Net.

3.1 Reconstruction Problem Overview: Formalizes LDCT degradation, objectives, and evaluation criteria (PSNR/SSIM, HU bias, NPS, EPI, ICC).

3.2 PGSR-Net: Architecture, loss design, training on Vietnamese NDCT with LDCT simulation, and ablations.

3.3 LRTR-Net: Lung-aware attention, multi-task learning with diagnostic priors, and cross-site robustness studies.

3.4 Experiments and Results: Datasets, protocols, baselines, quantitative/qualitative analysis, and clinical readouts.

Chapter 4 – Contrastive Learning for Domain-Robust Lung Cancer Prediction

Represents the second major contribution: CCP-Net and end-to-end screening pipeline.

4.1 Domain Adaptation in Medical Imaging: Problem setting, notation, and prior approaches.

4.2 CCP-Net: Dual-encoder contrastive alignment between NDCT and NDCT-like domains; fusion with radiomics; malignancy prediction head.

4.3 Evaluation and Comparative Analysis: AUC/F1, calibration, interpretability (Grad-CAM/embedding plots), and multi-center validation.

4.4 Discussion and Future Work: Clinical implications, limitations, and extensions to other organs/modalities.

3.3 Project Timeline

Table 1. *Project Timeline*

No.	Task	Result	Time
1	Conduct comprehensive literature review on CT physics, image reconstruction, and deep learning techniques.	Proposal	11/2025-4/2026
2	Review the proposal with the supervisor and revise accordingly.	Proposal	3/2026-4/2026
3	Defend the thesis outline before the academic committee.	Proposal	5/2026
4	Undertake research and compose the content for Chapter 1: Fundamentals of LDCT Reconstruction.	Chapter 1 Contents	8/2026-2/2027
5	Develop baseline models using traditional and CNN-based methods.	A scientific article	01/2027-6/2027
6	Submit the first scientific article to a peer-reviewed journal/conference.	Chapter 2 Contents	3/2027-9/2027
7	Conduct research for Chapter 2 and Chapter 3, focusing on hybrid, physics-informed, and self-supervised deep learning frameworks.	A scientific article	8/2027-2/2028
8	Implement and optimize proposed algorithms using public LDCT datasets.	Chapter 3 Contents	10/2027-6/2028
9	Submit two scientific papers based on novel methods and experimental results.	A scientific article	4/2027-10/2028
10	Undertake Chapter 4: Clinical Evaluation and Applications and Chapter 5: Conclusion and Future Work.	The thesis draft	10/2028-12/2029
11	Integrate methods into clinical data for validation.	An expert review	1/2028-2/2029
12	Write the first full draft of the thesis and submit for expert review.	Thesis Content	2/2028-4/2029
13	Organize a scientific seminar to present results.	Scientific reports	4/2028-7/2029
14	Revise the thesis based on feedback and defend at the grassroots level	Thesis Content	8/2028-11/2029
15	Finalize the thesis and defend it at the academic level	Thesis Content	

References

- [1] Rampinelli, C., Origgi, D. & Bellomi, M. Low-dose ct: technique, reading methods and image interpretation. *Cancer imaging* **12**, 548 (2013).
- [2] Kaur, R., Juneja, M. & Mandal, A. K. A comprehensive review of denoising techniques for abdominal ct images. *Multimedia Tools and Applications* **77**, 22735–22770 (2018).
- [3] Diwakar, M. & Kumar, M. A review on ct image noise and its denoising. *Biomedical Signal Processing and Control* **42**, 73–88 (2018).
- [4] Yang, L., Shangguan, H., Zhang, X., Wang, A. & Han, Z. High-frequency sensitive generative adversarial network for low-dose ct image denoising. *IEEE access* **8**, 930–943 (2019).
- [5] Shan, H. *et al.* Competitive performance of a modularized deep neural network compared to commercial algorithms for low-dose ct image reconstruction. *Nature Machine Intelligence* **1**, 269–276 (2019).
- [6] Du, W. *et al.* Stacked competitive networks for noise reduction in low-dose ct. *PloS one* **12**, e0190069 (2017).
- [7] Koetzier, L. R. *et al.* Deep learning image reconstruction for ct: technical principles and clinical prospects. *Radiology* **306**, e221257 (2023).
- [8] Gou, S. *et al.* Gradient regularized convolutional neural networks for low-dose ct image enhancement. *Physics in Medicine & Biology* **64**, 165017 (2019).
- [9] Chen, H. *et al.* Low-dose ct denoising with convolutional neural network. In *2017 IEEE 14th international symposium on biomedical imaging (ISBI 2017)*, 143–146 (IEEE, 2017).
- [10] Prior, F. *et al.* The public cancer radiology imaging collections of the cancer imaging archive. *Scientific data* **4**, 1–7 (2017).
- [11] Zeng, D. *et al.* A simple low-dose x-ray ct simulation from high-dose scan. *IEEE transactions on nuclear science* **62**, 2226–2233 (2015).
- [12] Manduca, A. *et al.* Projection space denoising with bilateral filtering and ct noise modeling for dose reduction in ct. *Medical physics* **36**, 4911–4919 (2009).

- [13] Chow, L. S. & Paramesran, R. Review of medical image quality assessment. *Biomedical signal processing and control* **27**, 145–154 (2016).
- [14] Gu, J. & Ye, J. C. Adain-based tunable cyclegan for efficient unsupervised low-dose ct denoising. *IEEE Transactions on Computational Imaging* **7**, 73–85 (2021).
- [15] Hu, X., Ding, X., Wu, R. & Zhang, M. Radiation dose of non-enhanced chest ct can be reduced 40% by using iterative reconstruction in image space. *Clinical radiology* **66**, 1023–1029 (2011).
- [16] Choi, K. Self-supervised learning for ct image denoising and reconstruction: a review. *Biomedical Engineering Letters* **14**, 1207–1220 (2024).
- [17] Eulig, E., Ommer, B. & Kachelrieß, M. Benchmarking deep learning-based low-dose ct image denoising algorithms. *Medical physics* **51**, 8776–8788 (2024).
- [18] Xu, R., Samat, A., Zhu, E., Li, E. & Li, W. Unsupervised domain adaptation with contrastive learning-based discriminative feature augmentation for rs image classification. *Remote Sensing* **16**, 1974 (2024).
- [19] Litjens, G. *et al.* A survey on deep learning in medical image analysis. *Medical image analysis* **42**, 60–88 (2017).
- [20] Yang, Q. *et al.* Big data from ct scanning. *JSM Biomed. Imag* **2**, 1003–1 (2015).
- [21] Ding, Q., Long, Y., Zhang, X. & Fessler, J. A. Statistical image reconstruction using mixed poisson-gaussian noise model for x-ray ct. *arXiv preprint arXiv:1801.09533* (2018).
- [22] Sukupova, L. Comparison of radiation dose and image quality for abdominal ct exams using photon-counting and energy-integrating ct: A self-controlled study including optimized patient positioning. *Radiography* **31**, 102909 (2025).
- [23] Stephens, K. Photon-counting ct enables lower contrast media for aortic imaging. *AXIS Imaging News* (2023).
- [24] Hara, A. K. *et al.* Iterative reconstruction technique for reducing body radiation dose at ct: feasibility study. *American Journal of Roentgenology* **193**, 764–771 (2009).

- [25] Jo, G. D. *et al.* 75% radiation dose reduction using deep learning reconstruction on low-dose chest ct. *BMC Medical Imaging* **23**, 121 (2023).
- [26] Nishikawa, M. *et al.* Image quality and radiologists' subjective acceptance using model-based iterative and deep learning reconstructions as adjuncts to ultrahigh-resolution ct in low-dose contrast-enhanced abdominopelvic ct: phantom and clinical pilot studies. *Abdominal Radiology* **47**, 891–902 (2022).
- [27] Pozzessere, C., von Garnier, C. & Beigelman-Aubry, C. Radiation exposure to low-dose computed tomography for lung cancer screening: should we be concerned? *Tomography* **9**, 166–177 (2023).
- [28] Goldman, L. W. Principles of ct: radiation dose and image quality. *Journal of nuclear medicine technology* **35**, 213–225 (2007).
- [29] Sahu, A. *et al.* Transforming ct imaging with deep learning: Noise reduction, artifact management, and clinical applications—a comprehensive review. *European Journal of Radiology Artificial Intelligence* 100042 (2025).
- [30] Mohammadinejad, P. *et al.* Ct noise-reduction methods for lower-dose scanning: strengths and weaknesses of iterative reconstruction algorithms and new techniques. *Radiographics* **41**, 1493–1508 (2021).
- [31] Szczykutowicz, T. P., Toia, G. V., Dhanantwari, A. & Nett, B. A review of deep learning ct reconstruction: concepts, limitations, and promise in clinical practice. *Current Radiology Reports* **10**, 101–115 (2022).
- [32] Jung, Y. *et al.* Radiation dose reduction using deep learning-based image reconstruction for a low-dose chest computed tomography protocol: a phantom study. *Quantitative Imaging in Medicine and Surgery* **13**, 1937 (2023).
- [33] Jiang, J.-M. *et al.* The value of deep learning image reconstruction in improving the quality of low-dose chest ct images. *Diagnostics* **12**, 2560 (2022).
- [34] Yang, S. *et al.* Low-dose ct denoising with a high-level feature refinement and dynamic convolution network. *Medical Physics* **50**, 3597–3611 (2023).
- [35] Zhang, W., Salmi, A., Yang, C. & Jiang, F. Innovative noise extraction and denoising in low-dose ct using a supervised deep learning framework. *Electronics* **13**, 3184 (2024).

- [36] Li, Y. *et al.* A phantom study comparing low-dose ct physical image quality from five different ct scanners. *Quantitative imaging in medicine and surgery* **12**, 766 (2022).
- [37] Kulathilake, K. S. H., Abdullah, N. A., Sabri, A. Q. M. & Lai, K. W. A review on deep learning approaches for low-dose computed tomography restoration. *Complex & Intelligent Systems* **9**, 2713–2745 (2023).
- [38] Ju, Y. H. *et al.* Ultra-low-dose lung screening ct with model-based iterative reconstruction: an assessment of image quality and lesion conspicuity. *Acta Radiologica* **59**, 553–559 (2018).
- [39] Moloney, F. *et al.* Low-dose ct imaging of the acute abdomen using model-based iterative reconstruction: a prospective study. *Emergency radiology* **26**, 169–177 (2019).
- [40] Kim, C. h. *et al.* The impact of deep learning reconstruction in low dose computed tomography on the evaluation of interstitial lung disease. *PloS one* **18**, e0291745 (2023).
- [41] Vardhanabhuti, V., Ilyas, S., Gutteridge, C., Freeman, S. J. & Roobottom, C. A. Comparison of image quality between filtered back-projection and the adaptive statistical and novel model-based iterative reconstruction techniques in abdominal ct for renal calculi. *Insights into imaging* **4**, 661–669 (2013).
- [42] Shuman, W. P. *et al.* Model-based iterative reconstruction versus adaptive statistical iterative reconstruction and filtered back projection in liver 64-mdct: focal lesion detection, lesion conspicuity, and image noise. *American journal of roentgenology* **200**, 1071–1076 (2013).
- [43] Boudabbous, S. *et al.* Model-based iterative reconstruction (mbir) for the reduction of metal artifacts on ct. *American Journal of Roentgenology* **205**, 380–385 (2015).
- [44] Pickhardt, P. J. *et al.* Abdominal ct with model-based iterative reconstruction (mbir): initial results of a prospective trial comparing ultralow-dose with standard-dose imaging. *American journal of roentgenology* **199**, 1266–1274 (2012).
- [45] Chen, L., Huang, Z., Long, Y. & Ravishankar, S. Enhancing low-dose ct image reconstruction by integrating supervised and unsupervised learning. *arXiv preprint arXiv:2311.12071* (2023).

- [46] Rahman, O. *et al.* Mbir training for a 2.5 d dl network in x-ray ct. *arXiv preprint arXiv:2309.13399* (2023).
- [47] Hsieh, J. Synthetization of high-dose images using low-dose ct scans. *Medical Physics* **51**, 113–125 (2024).
- [48] O'Regan, P. W. *et al.* Ultra-low-dose chest computed tomography with model-based iterative reconstruction in the analysis of solid pulmonary nodules: A prospective study. *World Journal of Radiology* **16**, 668 (2024).
- [49] Zhao, H., Qian, L., Zhu, Y. & Tian, D. Low dose ct image denoising: A comparative study of deep learning models and training strategies. *AI Medicine* 7–7 (2024).
- [50] Zhu, S. *et al.* Reduced-dose deep learning iterative reconstruction for abdominal computed tomography with low tube voltage and tube current. *BMC Medical Informatics and Decision Making* **24**, 389 (2024).
- [51] Sun, Q., He, N., Yang, P. & Zhao, X. Low dose computed tomography reconstruction with momentum-based frequency adjustment network. *Computer Methods and Programs in Biomedicine* **263**, 108673 (2025).
- [52] Wu, J. *et al.* Dual-domain deep prior guided sparse-view ct reconstruction with multi-scale fusion attention. *Scientific Reports* **15**, 16894 (2025).
- [53] Li, X., Zhang, L., Yang, J. & Teng, F. Role of artificial intelligence in medical image analysis: A review of current trends and future directions. *Journal of Medical and Biological Engineering* **44**, 231–243 (2024).
- [54] Zhang, S. *et al.* Effectiveness of ai for enhancing computed tomography image quality and radiation protection in radiology: Systematic review and meta-analysis. *Journal of Medical Internet Research* **27**, e66622 (2025).
- [55] Suganeshwari, G. *et al.* Dtbv: A deep transfer-based bone cancer diagnosis system using vgg16 feature extraction. *Diagnostics* **13**, 757 (2023).
- [56] DeCunha, J. M., Newpower, M. & Mohan, R. Gpu-accelerated calculation of proton microdosimetric spectra as a function of target size, proton energy, and bounding volume size. *Physics in Medicine & Biology* **68**, 165012 (2023).

- [57] Han, M. & Baek, J. Direct estimation of the noise power spectrum from patient data to generate synthesized ct noise for denoising network training. *Medical physics* **51**, 1637–1652 (2024).
- [58] Wang, Z. Image quality assessment: Form error visibility to structural similarity. *IEEE Trans. Image Process.* **13**, 604–606 (2004).
- [59] Kang, E., Min, J. & Ye, J. C. A deep convolutional neural network using directional wavelets for low-dose x-ray ct reconstruction. *Medical physics* **44**, e360–e375 (2017).